Construction of Maiorana-McFarland type cryptographically significant Boolean functions with good implementation properties

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1 Introduction

Boolean functions (for details, see [Car10, CS09]) are fundamental to both computing and communication technologies. Among them, bent functions [Dil74, Mes16, Rot76] form a particularly notable subclass due to their rich combinatorial structure and wide-ranging applications in coding theory and cryptology. A well-known subclass of bent functions, the Maiorana-McFarland functions, abbreviated as MM functions, has been extensively used as a basic building block in the explicit construction of various combinatorial objects, particularly in designing Boolean functions with desirable cryptographic properties, especially in the context of symmetric cipher design.

To proceed with the technical discussion, we begin by describing the class of MM functions defined on $n = 2k$ variables. The input variables are partitioned into two disjoint sets: $\mathbf{x} = (x_1, x_2, \ldots, x_k)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_k)$. An MM function is then given by the expression

$$f(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \pi(\mathbf{y}) \oplus g(\mathbf{y}),$$

where $\pi : \mathbb{F}_2^k \to \mathbb{F}_2^k$ is a permutation function (i.e., a bijective mapping on k-bit vectors, which can be viewed as a special class of vectorial Boolean functions), and $g : \mathbb{F}_2^k \to \mathbb{F}_2$ is a scalar Boolean function with k inputs. How many such functions are there? One can see that there are $(2^k)!$ such permutations and 2^{2^k} possible ways to choose $g(\mathbf{y})$. Thus, the total number of such functions is $(2^{\frac{n}{2}})! \cdot 2^{2^{\frac{n}{2}}}$, for even $n = 2k$. This is indeed a huge class of functions. By using a simple counting argument, one can show that circuit realization of most of them requires exponential depth circuits on a standard gate basis. On the other hand, it is to be noted that there are many functions that can be implemented with a significantly small depth circuit. For example, if we choose π as the identity permutation and g as the identity 0, then the function can be implemented using a significantly small amount of circuit components. The function is then simply $x_1y_1 \oplus x_2y_2 \oplus \cdots \oplus x_ky_k$. In fact, it can be implemented with k two-input AND gates placed in parallel and $k-1$ two-input XOR gates placed in the form of a complete binary tree with $\lceil \log_2 k \rceil$ depth. At this point, it is quite tempting to present another view of the MM bent functions. Suppose that the left half of the input variables is $\mathbf{y}$ and the right half is $\mathbf{x}$. Linear functions are

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depicted by $\bigoplus_{i=1}^{k} a_i x_i$, for all 2^k different options of $(a_1, a_2, \ldots, a_k)$. Each such linear function has a truth table of length 2^k, and concatenating different 2^k of them, one can obtain a truth table of length 2^{2k} for a n-variable function, where $n = 2k$. In a generalized framework, the small truth tables of the linear functions can be permuted among each other, which is defined by π. Moreover, whether the linear function will be presented as is or complemented will be decided by the function g. It can be proved that such functions are bent, i.e., they have nonlinearity $2^{n-1} - 2^{\frac{n}{2}-1}$.

Coming back to our original discussion, bent functions are generally not directly employed as cryptographic primitives since they are not balanced. However, it is important to note that, from a global perspective, these functions exhibit the best possible spectral properties. Their Walsh-Hadamard spectra (which relate to confusion and resistance against linear cryptanalysis) and autocorrelation spectra (which relate to diffusion and resistance against differential cryptanalysis) are provably optimal. Modified versions of bent functions typically sacrifice some nonlinearity and autocorrelation strength, but they can achieve other desirable cryptographic properties such as balancedness and resiliency. One well-known modification technique involves replacing the truth table of the all-zero linear function (i.e., $\bigoplus_{i=1}^{k} a_i x_i$ with all $a_i = 0$) by a nonlinear balanced function on $k = \frac{n}{2}$ variables in the construction of MM bent functions, as described in [Dob94]. This technique yields balanced functions on even n variables, with nonlinearity $2^{n-1} - 2^{\frac{n}{2}} + \mathrm{nlb}(\frac{n}{2})$, where $\mathrm{nlb}(t)$ denotes the maximum nonlinearity of a balanced Boolean function in t variables. Over the decades, modifications of MM bent functions in this spirit have remained central to the construction of cryptographic Boolean functions. We refer to the resulting functions and constructions as MM-type functions and MM-type constructions, respectively. A substantial body of work has focused on building concrete cryptographic Boolean functions following this approach; see, for example, the recent studies in [KMT19, TM18, TKMM19, TMM21].

We emphasize that, the primary focus in the area of constructing cryptographic Boolean functions has consistently been on improving cryptographic parameters. To the best of our knowledge, little attention has been given to the circuit realization cost of implementing these candidate functions. Although some works have implicitly considered this aspect, there has been no dedicated or systematic study addressing this issue. In this work, we initiate a study and provide a significant formalization of a methodology for constructing functions that achieve small size circuit realization.

1.1 Related Works

In fact, efficient circuit realization of Boolean functions have been of interest in some earlier works, but the number of such works is limited. In this regard, one may refer to [SM01, SM03], where the circuits for cryptographically significant functions have been considered. Later in [GS04], efficient representation and software implementation of resilient Maiorana-McFarland type S-boxes have been studied. However, in these works, they did not refer to how one can realize functions on a large number of variables having small size circuits and achieving very good nonlinearity as well as autocorrelation properties. Very recently, certain evolutionary algorithms have been exploited in [OKK23] to construct functions with good parameters, and that improves certain results of [KMT19]. However, such search methodologies can only work up to a certain number of variables, namely 26, as pointed out in [OKK23]. The question of such efficient circuit realization was raised a few years back in [CMY+18], too. Very recently, such issues have also been discussed in [TKMM19], where authors have implicitly considered efficient circuit realization as one of the design criteria for their works.

1.2 Organization & Contribution

The foundation of our construction is presented in Section 2, where we revisit several recent works on MM-type constructions, specifically [KMT19, TM18, TKMM19]. While these studies explore important cryptographic parameters, it is important to emphasize that none of them provide explicit constructions of Boolean functions that are efficiently realizable on circuits for arbitrarily large numbers of variables. In particular, [TKMM19, Section 4.3] highlights implementation challenges, noting that the logic gate count in their construction grows exponentially with the number of input variables. This growth is primarily due to reliance on decoder-based strategies, which require $O(2^k)$ logic gates.

In contrast, we approach this challenge from a different perspective. Our aim is to design MM-type Boolean functions that preserve strong cryptographic properties while enabling realizations with significantly lower circuit complexity. We propose new strategies that yield functions implementable using only a small number of logic gates, thus offering a practical alternative to existing methods.

Section 3 presents an explicit construction of balanced Boolean functions that achieve high nonlinearity and very low absolute autocorrelation. Additionally, the construction allows flexibility in tuning the algebraic degree to a desired level. To the best of our knowledge, this is the first explicit construction in the MM framework that simultaneously satisfies key cryptographic criteria and supports compact circuit realizations.

More precisely, for $n = 4t \geq 20$ with even t, the nonlinearity of the constructed Boolean function $f \in \mathcal{B}_n$ from Construction 1 is given by

$$nl(f) = 2^{n-1} - 2^{\frac{n}{2}-1} - 3 \cdot 2^{\frac{n}{4}-1} - 2^{\frac{n}{8}}.$$

When t is odd, the nonlinearity satisfies

$$2^{n-1} - 2^{\frac{n}{2}-1} - 3 \cdot 2^{\frac{n}{4}-1} - 2^{\frac{n-4}{8}} \leq nl(f) \leq 2^{n-1} - 2^{\frac{n}{2}-1} - 3 \cdot 2^{\frac{n}{4}-1}.$$

Moreover, the absolute indicator Δ_f of f satisfies

$$\Delta_f \leq 2^{\frac{n}{2}} - 2^{\frac{n}{2}-2} + 9 \cdot 2^{\frac{n}{4}}.$$

These parameters are strictly better than those obtained in [TKMM19, Theorem 13] for functions with $n = 2k = 4t \geq 20$.

Importantly, our construction supports compact circuit realizations. In fact the function can be realized using at most $6n$ logic gates (see Section 3.4 for detailed analysis), highlighting its practical feasibility for use in resource-constrained cryptographic applications.

Finally, in Section 4, we explore potential applications of our construction in two important domains: fully homomorphic encryption (FHE) schemes and symmetric key primitives resistant to differential fault attacks (DFA). We illustrate how our function can replace the core Boolean component in two well-known stream cipher architectures, resulting in designs that offer enhanced cryptographic properties while maintaining compact circuit realizations. These results demonstrate promising directions for future practical integration and deployment. Section 5 concludes the paper.

We must present the caveat that our examples in Section 4 are not sufficient to claim that these functions are implementation-efficient or fully resistant against DFA. We provide certain improvements towards the resistance against the previous DFA attacks on two example ciphers, namely FLIP and Grain v1, but we do not claim that such new design proposals with our proposed functions are fully resistant against DFA. Further investigations are required towards actual hardware implementation of our proposed functions as in Section 3 and to plug them in a concrete design of a stream cipher.

1.3 Preliminaries

First, we present some basic definitions and notations related to Boolean functions. Let $\mathbb{F}_2$ denote the finite field of two elements (i.e., the field of characteristic 2), and let $\mathbb{F}_{2^n}$ be its degree-n extension. The n-dimensional vector space over $\mathbb{F}_2$ is denoted by $\mathbb{F}_2^n$, and its elements are represented as $\mathbf{x} = (x_1, x_2, \dots, x_n)$, where each $x_i \in \mathbb{F}_2$ for $1 \le i \le n$. The set of nonzero elements in $\mathbb{F}_2^n$ is denoted by $\mathbb{F}_2^{n*}$. The *Hamming weight* of a vector $\mathbf{x} \in \mathbb{F}_2^n$ is defined as

$$wt(\mathbf{x}) = \sum_{i=1}^{n} x_i,$$

where the sum is over the integers. For a finite set S, the cardinality (number of elements) is denoted by $|S|$. A function from $\mathbb{F}_2^n$ to $\mathbb{F}_2^m$ is called an n-variable Boolean function with m outputs. In most cases, we consider $m = 1$, and denote the set of all such Boolean functions as $\mathcal{B}_n$. Any function $f \in \mathcal{B}_n$ can be expressed as a multivariate polynomial over $\mathbb{F}_2$ in the following form, known as the *algebraic normal form* (ANF):

$$f(\mathbf{x}) = \bigoplus_{\mathbf{a} \in \mathbb{F}_2^n} \mu_{\mathbf{a}} x_1^{a_1} x_2^{a_2} \cdots x_n^{a_n},$$

where $\mu_{\mathbf{a}} \in \mathbb{F}_2$ for all $\mathbf{a} \in \mathbb{F}_2^n$. The *algebraic degree* of f is defined as

$$\deg(f) = \max\{wt(\mathbf{a}) : \mu_{\mathbf{a}} \neq 0\}.$$

If $\deg(f) \le 1$, the function is called *affine*; if, in addition, the constant term is zero, it is called *linear*. The *support* of a Boolean function f is defined as

$$\mathrm{supp}(f) = \{\mathbf{x} \in \mathbb{F}_2^n : f(\mathbf{x}) = 1\},$$

and the cardinality $|\mathrm{supp}(f)|$ is referred to as the *weight* of f. A function f is said to be *balanced* if $|\mathrm{supp}(f)| = 2^{n-1}$. The *Walsh–Hadamard transform* of a Boolean function $f \in \mathcal{B}_n$ at a point $\mathbf{a} \in \mathbb{F}_2^n$ is given by:

$$W_f(\mathbf{a}) = \sum_{\mathbf{x} \in \mathbb{F}_2^n} (-1)^{f(\mathbf{x}) \oplus \mathbf{a} \cdot \mathbf{x}}.$$

The multiset $\{W_f(\mathbf{a}) : \mathbf{a} \in \mathbb{F}_2^n\}$ is called the *Walsh–Hadamard spectrum* of f. By Parseval's identity, we have:

$$\sum_{\mathbf{a} \in \mathbb{F}_2^n} W_f^2(\mathbf{a}) = 2^{2n},$$

which implies that $\max_{\mathbf{a} \in \mathbb{F}_2^n} |W_f(\mathbf{a})| \ge 2^{n/2}$ for any Boolean function f. The *nonlinearity* of $f \in \mathcal{B}_n$ is defined as the minimum Hamming distance between f and all affine functions on n variables. In terms of the Walsh–Hadamard transform, it can be expressed as:

$$nl(f) = 2^{n-1} - \frac{1}{2} \max_{\mathbf{a} \in \mathbb{F}_2^n} |W_f(\mathbf{a})|.$$

It is known that the maximum nonlinearity for any Boolean function in n variables is upper-bounded by $2^{n-1} - 2^{n/2-1}$. A Boolean function that achieves this bound is called a *bent function*. Therefore, for any bent function $f \in \mathcal{B}_n$, we have:

$$nl(f) = 2^{n-1} - 2^{\frac{n}{2}-1}, \quad \text{and} \quad |W_f(\mathbf{a})| = 2^{n/2} \text{ for all } \mathbf{a} \in \mathbb{F}_2^n.$$

Consequently, bent functions exist only for even values of n, and they are not balanced.

Another important cryptographic property is the *autocorrelation* of $f \in \mathcal{B}_n$ at a point $\mathbf{a} \in \mathbb{F}_2^n$, defined as:

$$C_f(\mathbf{a}) = \sum_{\mathbf{x} \in \mathbb{F}_2^n} (-1)^{f(\mathbf{x}) \oplus f(\mathbf{x} \oplus \mathbf{a})}.$$

For any bent function $f \in \mathcal{B}_n$, we have $C_f(\mathbf{a}) = 0$ for all $\mathbf{a} \neq \mathbf{0}$, and $C_f(\mathbf{0}) = 2^n$. Finally *absolute indicator* of f is given by:

$$\Delta_f = \max_{\mathbf{a} \in \mathbb{F}_2^{n*}} |C_f(\mathbf{a})|.$$

From a cryptographic point of view, one needs to construct a balanced Boolean function having very high nonlinearity and very low absolute indicator to resist the linear and differential attacks on symmetric ciphers.

Throughout this work, only these standard logic gates (AND, OR, NOT, XOR) with fan-in 2 will be used for circuit realization of logic functions. A 2×1 multiplexer (MUX) is a digital circuit that selects one of two input signals based on the value of a single select line and forwards it to the output. It has two data inputs (I_0 and I_1), one select input (S), and one output (Y). In total, the 2×1 MUX requires four logic gates.

2 Efficient circuit realization strategy for notable MM-Type constructions

In 1994, Dobbertin [Dob94] constructed balanced Boolean functions on even number of variables, $2k$, by modifying the all-zero linear function block (recall our earlier discussion on how the truth table of an MM function can be interpreted as the concatenation of linear functions) of an MM function. This was achieved by completely replacing it with balanced functions in k variables that exhibit high nonlinearity. Building on the principles of this work, balanced Boolean functions with very good autocorrelation spectrum and nonlinearity were recently constructed in [KMT19, TM18, TKMM19].

In [KMT19] and [TM18], Kavut et al. and Tang et al., respectively, modified two fixed positions in each linear function block by substituting them with two nonlinear balanced functions possessing specific properties. Later, in [TKMM19], Tang et al. further refined this approach by replacing the first all-zero linear function block with a nonlinear function in k variables and substituting the all-zero inputs of all other linear function blocks with another nonlinear function in k variables. These two smaller sub-functions were designed to satisfy certain desirable cryptographic properties.

In the following part of this section, we revisit these constructions from the perspective of circuit realization and identify the key bottlenecks that hinder the synthesis of small sized circuit for them .

2.1 The case of Balanced Boolean functions, with high nonlinearity given in [Dob94]

Let us recall the structure of MM functions defined on $n = 2k$ variables. The input variables are partitioned into two sets: $\mathbf{x} = (x_1, x_2, \ldots, x_k)$ and $\mathbf{y} = (y_1, y_2, \ldots, y_k)$. Then MM function is defined as $h(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \pi(\mathbf{y}) \oplus g(\mathbf{y})$, where π is a permutation from $\mathbb{F}_2^k$ to $\mathbb{F}_2^k$, and g is a Boolean function in $\mathcal{B}_k$. Now, suppose $g(\mathbf{y}) = 0$ for all $\mathbf{y} \in \mathbb{F}_2^k$, and the all-zero linear function block is replaced with a balanced Boolean function on k variables. Then it is straightforward to verify that the resulting function is a balanced Boolean function on $n = 2k$ variables. The outputs of the circuit shown in Fig. 1 correspond to such a balanced Boolean function constructed in this manner. In [Dob94], Dobbertin conjectured that substituting the all-zero linear function block with a balanced Boolean function of maximum nonlinearity on k variables yields a balanced Boolean function of maximum nonlinearity on $n = 2k$ variables. To date, this remains a major open problem in the literature. In this work, however, we approach this constructions from the perspective of circuit realization and focus on the feasibility of realizing such functions efficiently.

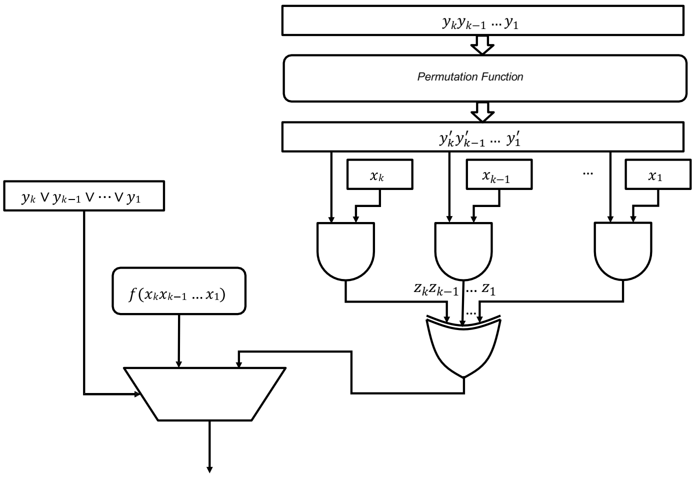

Figure 1: Circuit realization of Dobbertin's construction [Dob94]

The circuit realization cost of such functions depends primarily on two components: the permutation π over $\mathbb{F}_2^k$ and the nonlinear balanced Boolean function f on k variables used to replace the all-zero linear function block. We begin by noting that there exist permutations over $\mathbb{F}_2^k$ that admit small-sized circuit realizations. In particular, any invertible $k \times k$ binary matrix defines a permutation on $\mathbb{F}_2^k$ and can be implemented using $O(k^2)$ logic gates. Assuming, for the moment, that the balanced Boolean function f on k variables can also be realized by a small-sized circuit, it follows that the overall construction can be implemented efficiently using small size circuit. Moreover, when k is a power of 2, the last assumption on f can be lifted by adopting a recursive construction. As one may start with $k = 4$, where balanced Boolean function with the highest nonlinearity functions are known and recursively apply the same construction to obtain a balanced function f on $2k$ variables. Consequently, for such values of k, the Boolean function conjectured by Dobbertin can be realized using only a quadratic number of logic gates with respect to the total number of input variables.

Furthermore, it is worth noting that several nonlinear permutations admit compact circuit realizations. For instance, consider the permutation

$$\pi(y_k, \ldots, y_2, y_1) = (y_k, \ldots, y_2, y_k y_{k-1} \cdots y_2 \oplus y_1)$$

for $k \geq 3$. This permutation can be implemented using at most $k - 2$ AND gates and a single XOR gate. This is a valuable insight, as it allows for adjusting the algebraic degree of the resulting function, as elaborated in the following remark.

Remark 1. It is important to highlight that the algebraic degree of the resulting functions can be made high, an essential requirement for cryptographic primitives in nonlinear combiners or filter generator designs that incorporate Linear Feedback Shift Registers (LFSRs) or Nonlinear Feedback Shift Registers (NFSRs). In general, the algebraic degree of the balanced Boolean functions constructed via the methods in [Dob94, KMT19, TM18, TKMM19] depends on the degrees of the underlying permutation and the component functions. As long as these components possess high algebraic degree, the overall construction inherits this desirable property.

2.2 The case of Balanced Boolean functions with high nonlinearity and low absolute indicators given in [KMT19, TM18, TKMM19]

While the functions constructed in [Dob94] achieve good nonlinearity, they generally do not exhibit low absolute indicators. To address this, Kavut et al. [KMT19] and Tang et al. [TM18, TKMM19] proposed constructions of balanced Boolean functions by modifying MM bent functions to improve both the autocorrelation spectrum and nonlinearity. Instead of altering a single linear function block in the truth table of the MM bent function, their approaches incorporated two distinct nonlinear sub-functions defined over smaller variable sets. Specifically, in [KMT19] and [TM18], the construction involved modifying the outputs of the MM function at two fixed positions within each linear function block using carefully selected sub-functions. In contrast, [TKMM19] focused on altering the outputs corresponding to the first all-zero input of each linear function block and the all-zero blocks themselves, again using two tailored sub-functions.

From a circuit realization perspective, if one adopts the decoder-based view discussed in the earlier section of this paper, the total number of logic gates required to implement such balanced Boolean functions grows exponentially with k, i.e., on the order of $\mathcal{O}(k2^k)$. However, as demonstrated below, these functions can be realized more efficiently by following the design illustrated in Fig. 2, with the addition of a single 2×1 multiplexer. This alternative approach significantly reduces the realization cost by linking it closely to the chosen permutation π and the specific sub-functions employed.

Before delving into the technical specifics of this construction, we present a schematic overview of the circuit, which is aligned with the conceptual framework depicted in Fig. 1. Let π be a permutation over $\mathbb{F}_2^k$ satisfying $\pi(\mathbf{0}) = \mathbf{0}$. Let $f_1, f_2 \in \mathcal{B}_k$ be Boolean functions with desirable cryptographic properties, and let $h_1, h_2 \in \mathcal{B}_k$ be atomic indicator functions used to selectively modify the outputs of the function at two fixed positions. The circuit shown in Fig. 2 then generates the balanced Boolean functions in $2k$ variables constructed in [KMT19, TM18, TKMM19], depending on the choice of f_i and h_i for $i = 1, 2$.

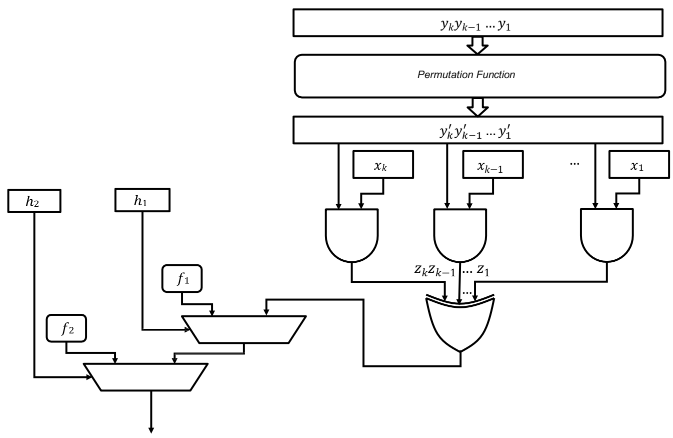

Figure 2: Construction of balanced Boolean function proposed in [KMT19, TM18, TKMM19]

The outputs of the 2×1 multiplexers are determined by h_1 and h_2, which can be implemented using only 4 logic gates. Therefore, the total logic gate count for circuit realization of Fig. 2 is primarily influenced by the circuit size of the permutation π and the sub-functions f_1 and f_2. We will now discuss these cases individually, providing concrete examples for small values of k.

2.2.1 Circuits of Boolean functions given in [KMT19, TM18]

In [KMT19, TM18], balanced Boolean functions in $2k$ variables (with odd $k \geq 9$ in [TM18] and even $k \geq 10$ in [KMT19]) exhibiting balancedness and good autocorrelation properties were constructed by modifying a certain class of bent function (known as Partial Spread funtions), of the form $\mathrm{Tr}_1^k\left(\frac{\lambda x}{y}\right)$ for all $x, y, \lambda \in \mathbb{F}_{2^k}$, with $\lambda \neq 0$ where $\mathrm{Tr}_1^k(x) = x \oplus x^2 \oplus \cdots \oplus x^{2^{k-1}}$. The construction also involves two Boolean sub-functions f_1 and f_2 in k variables such that $wt(f_1) + wt(f_2) = 2^k$. The outputs of $\mathrm{Tr}_1^k\left(\frac{\lambda x}{y}\right)$ are modified as follows: by $f_1(y)$ when $x = 0$ and by $f_2(y)$ when $x = \mu$, where $\mu \in \mathbb{F}_{2^k} \setminus \{0\}$. Consequently, the balanced Boolean function f in $2k$ variables (with $2k \geq 18$) is defined as:

$$f(x, y) = \begin{cases} f_1(y), & \text{if } x = 0, \\ f_2(y), & \text{if } x = \mu, \\ \mathrm{Tr}_1^k\left(\frac{\lambda x}{y}\right), & \text{if } x \neq 0 \text{ and } x \neq \mu, \end{cases}$$

While the condition $wt(f_1) + wt(f_2) = 2^k$ ensures that the resulting function is balanced, it does not guarantee good autocorrelation spectrum. Further criteria are required to achieve desirable properties as given in [TM18, Lemma 6].

From a circuit realization perspective, several key observations can be made regarding these constructions. Firstly, in [TKMM19, Theorem 2], it was shown that the Boolean function $\mathrm{Tr}_1^k\left(\frac{\lambda x}{y}\right)$ can be expressed as a concatenation of 2^k linear functions, using a specific nonlinear permutation $y \mapsto y^{2^k-2}$ over the finite field $\mathbb{F}_{2^k}$. Secondly, the function corresponding to $\mathrm{Tr}_1^k(wx)$, where $w = y^{2^k-2}$ is fixed when y is fixed and $\lambda = 1$, is linear in k variables over the vector space $\mathbb{F}_2^k \times \mathbb{F}_2^k$. Lastly, in these constructions, each linear function is modified at two fixed points. Assuming $\lambda = 1$, the circuit realisation cost of the function in question, when realized by following Fig. 2, depends on both the permutation $y \mapsto y^{2^k-2}$ and the smaller functions f_1 and f_2. It is important to note that the efficient implementation of the permutation $y \mapsto y^{2^k-2}$ for arbitrary k is not yet determined. Similarly, it remains unclear whether the functions f_1 and f_2 can always be implemented with small circuits for all values of k.

2.2.2 Circuits of Boolean functions given in [TKMM19]

Tang et al. [TKMM19] constructed balanced Boolean functions in $2k$ variables with very high nonlinearity and very low absolute indicator (maximum absolute value of the autocorrelation spectrum) by modifying a simple MM bent function of the form $\mathbf{x} \cdot \pi(\mathbf{y})$, where π is a permutation over $\mathbb{F}_2^k$ with $\pi(\mathbf{0}) = 0$. They utilized two Boolean functions, f_1 and f_2, in k variables, ensuring that $f_1(\mathbf{0}) = f_2(\mathbf{0}) = 0$ and that $wt(f_1) + wt(f_2) = 2^{k-1}$. The output of $\mathbf{x} \cdot \pi(\mathbf{y})$ is modified by $f_1(\mathbf{y})$ when $\mathbf{x} = \mathbf{0}$, and by $f_2(\mathbf{x})$ when $\mathbf{y} = \mathbf{0}$. The resulting balanced Boolean function f in $2k \geq 4$ variables is defined as:

$$f(\mathbf{x}, \mathbf{y}) = \begin{cases} f_1(\mathbf{y}), & \text{if } (\mathbf{x}, \mathbf{y}) \in \{\mathbf{0}\} \times \mathbb{F}_2^{k*}, \\ f_2(\mathbf{x}), & \text{if } (\mathbf{x}, \mathbf{y}) \in \mathbb{F}_2^{k*} \times \{\mathbf{0}\}, \\ \mathbf{x} \cdot \pi(\mathbf{y}), & \text{Otherwise} \end{cases}$$

where $\mathbb{F}_2^{k*} = \mathbb{F}_2^k \setminus \{\mathbf{0}\}$, and π, f_1, and f_2 are defined as above. For other cryptographic properties (specifically nonlinearity and absolute indicators) of balanced function f, the functions f_1 and f_2 should satisfy some additional criterion, as described below.

As defined in [TKMM19, Definition 8], f_1 needs to satisfy

$$|W_{f_1}(\mathbf{a})| \leq \begin{cases} 2^{k-1} + 2^{\frac{k}{2}-1}, & \text{if } \mathbf{a} = \mathbf{0}, \\ 3 \cdot 2^{\frac{k}{2}}, & \text{if } \mathbf{a} \in \mathbb{F}_2^k \setminus \{\mathbf{0}\}, \end{cases}$$

and

$$|C_{f_1}(\mathbf{a})| \geq \begin{cases} 2^{k}, & \text{if } \mathbf{a} = \mathbf{0}, \\ 2^{k-2}, & \text{if } \mathbf{a} \in \mathbb{F}_2^k \setminus \{\mathbf{0}\}. \end{cases}$$

As defined in [TKMM19, Theorem 12], f_2 needs to satisfy

$$|W_{f_2}(\mathbf{a})| \leq \begin{cases} 2^{k-1} + 3 \cdot 2^{\frac{k}{2}-1}, & \text{if } \mathbf{a} = \mathbf{0}, \\ 5 \cdot 2^{\frac{k}{2}-1}, & \text{if } \mathbf{a} \in \mathbb{F}_2^k \setminus \{\mathbf{0}\}, \end{cases}$$

and

$$|C_{f_2}(\mathbf{a})| \geq \begin{cases} 2^{k}, & \text{if } \mathbf{a} = \mathbf{0}, \\ 2^{k-2} - 2^{\frac{k}{2}+2}, & \text{if } \mathbf{a} \in \mathbb{F}_2^k \setminus \{\mathbf{0}\}. \end{cases}$$

As one can see directly from the construction, a key advantage of this construction is that it operates on any permutation e.g. even on identity permutation which can be realized without actually using any logic gates. As discussed earlier, one can use nonlinear permutations with high algebraic degree for this construction. However the main bottleneck in terms of circuit size lies in the existence of small-size circuits for f_1 and f_2. To the best of our understanding, for larger values of k, the existence of small-sized circuits for f_1 and f_2, satisfying criteria as described in [TKMM19, Theorem 12] and [TKMM19, Theorem 8] are not immediately apparent.

2.3 Efficient circuit realizability as a design criteria

This work introduces efficient circuit realizability as a critical design criterion. From the discussions in earlier portion of this section, we identify the key bottlenecks that hinder small size circuit realization. More specifically, we identify two main challenges for efficient circuit realization: the permutation function and the component functions used to modify linear function blocks. While many MM-type constructions allow choosing the permutation realizable in small size circuit, sub-functions that modify the MM function at different positions of its truth table are difficult to implement in small circuits, particularly for large numbers of input variables. Keeping this in mind, in the next section, we present a construction of a balanced Boolean function for $n = 4t$ that exhibits very high nonlinearity and a low absolute indicator, while still being realizable in small size circuits.

3 A class of ultra-lightweight Boolean functions with strong cryptographic properties

In earlier sections, we presented efficient implementation strategies for MM-type functions. While these functions can be implemented for a small number of variables, the implementation for larger input sizes is not straightforward. In this section, we propose a similar construction that leverages the circuit realization strategy outlined earlier. Specifically, we introduce a construction with two component functions, which have a provably low circuit realization cost in an asymptotic sense. It is to be underlined again that the constructed balanced Boolean function exhibits high algebraic degree, very good nonlinearity and autocorrelation spectra.

3.1 Two ultra-lightweight sub-functions for main construction

We first define a balanced Boolean function in $t \geq 5$ variables, which is a concatenation of certain bent functions.

Definition 1. Let t be an integer such that $t \geq 5$. Let g be a t-variable Boolean function defined as follows: $g(\mathbf{r}) = \bigoplus_{i=1}^{\iota} r_i r_{i+\iota} \oplus \bigoplus_{i=2\iota+1}^{t} r_i$, where $\mathbf{r} = (r_1, r_2, \ldots, r_t) \in \mathbb{F}_2^t$ and $\iota = \lfloor \frac{t-1}{2} \rfloor$.

Here, $\oplus_{i=1}^{\iota} r_i r_{i+\iota}$ is a bent function in 2ι variables. If t is odd, then $\iota = \frac{t-1}{2}$, so the function g is a concatenation of truth tables of two bent functions in $t-1$ variables, a bent and its complementary. If t is even, then $\iota = \frac{t}{2} - 1$, and so, the function g is a concatenation of four bent functions in $t-2$ variables of the form

$$\oplus_{i=1}^{\iota} r_i r_{i+\iota} || \oplus_{i=1}^{\iota} r_i r_{i+\iota} \oplus 1 || \oplus_{i=1}^{\iota} r_i r_{i+\iota} \oplus 1 || \oplus_{i=1}^{\iota} r_i r_{i+\iota}.$$

Next we will move to derive some cryptographic properties of g. It is easy to see that g is a balanced function in t variables. Next two lemmas show that g has high nonlinearity and low absolute indicators. The proof of both of the lemmas rely heavily on symmetry, modularity of $\mathbb{F}_2^t$ and structured cancellation within the exponential terms of the Walsh-Hadamard transform, where $W_g(\mathbf{d})$ provides the Walsh transform value of the function g at the point $\mathbf{d}$.

Lemma 1. *Let $t \geq 5$ be an integer and $g \in \mathcal{B}_t$ be the function defined in Definition 1. Then for any $\mathbf{d} = (d_1, d_2, \cdots, d_t) \in \mathbb{F}_2^t$, if t is even, we have*

$$W_g(\mathbf{d}) = \begin{cases} 0, & \text{if } (d_t, d_{t-1}) \in \{(0,0),(0,1),(1,0)\} \\ (-1)^{\bigoplus_{i=1}^{\iota} d_i d_{i+\iota}} \cdot 2^{\frac{t}{2}+1}, & \text{if } (d_t, d_{t-1}) = (1,1) \end{cases},$$

and if t is odd, we have

$$W_g(\mathbf{d}) = \begin{cases} 0, & \text{if } d_t = 0 \\ (-1)^{\bigoplus_{i=1}^{\iota} d_i d_{i+\iota}} \cdot 2^{\frac{t+1}{2}}, & \text{if } d_t = 1 \end{cases}$$

where $\iota = \lfloor \frac{t-1}{2} \rfloor$.

Proof. Let $g_1(\mathbf{x}) = \oplus_{i=1}^{\iota} x_i x_{i+\iota}$, where $\mathbf{x} \in \mathbb{F}_2^{2\iota}$, and t be an even integer. Then $\iota = \frac{t}{2} - 1$ and $g(\mathbf{r}) = \oplus_{i=1}^{\iota} r_i r_{i+\iota} \oplus r_{t-1} \oplus r_t$. For any $\mathbf{d} = (\mathbf{d}', d_{t-1}, d_t) \in \mathbb{F}_2^{t-2} \times \mathbb{F}_2 \times \mathbb{F}_2$, we have

$$\begin{aligned} W_g(\mathbf{d}) &= \sum_{(\mathbf{r}', r_{t-1}, r_t) \in \mathbb{F}_2^{t-2} \times \mathbb{F}_2 \times \mathbb{F}_2} (-1)^{g(\mathbf{r}', r_{t-1}, r_t) \oplus \mathbf{d}' \cdot \mathbf{r}' \oplus d_{t-1} r_{t-1} \oplus d_t r_t} \\ &= \sum_{\mathbf{r}' \in \mathbb{F}_2^{t-2}} (-1)^{g_1(\mathbf{r}') \oplus \mathbf{d}' \cdot \mathbf{r}'} + \sum_{\mathbf{r}' \in \mathbb{F}_2^{t-2}} (-1)^{g_1(\mathbf{r}') \oplus 1 \oplus \mathbf{d}' \cdot \mathbf{r}' \oplus d_{t-1}} \\ &\quad + \sum_{\mathbf{r}' \in \mathbb{F}_2^{t-2}} (-1)^{g_1(\mathbf{r}') \oplus 1 \oplus \mathbf{d}' \cdot \mathbf{r}' \oplus d_t} + \sum_{\mathbf{r}' \in \mathbb{F}_2^{t-2}} (-1)^{g_1(\mathbf{r}') \oplus \mathbf{d}' \cdot \mathbf{r}' \oplus d_{t-1} \oplus d_t} \\ &= \left(1 + (-1)^{d_{t-1} \oplus 1} + (-1)^{d_t \oplus 1} + (-1)^{d_{t-1} \oplus d_t}\right) W_{g_1}(\mathbf{d}') \\ &= \begin{cases} 0, & \text{if } (d_t, d_{t-1}) \in \{(0,0),(0,1),(1,0)\} \\ 2^{\frac{t}{2}+1} (-1)^{\bigoplus_{i=1}^{\iota} d_i d_{i+\iota}}, & \text{if } (d_t, d_{t-1}) = (1,1) \end{cases}. \end{aligned}$$

Suppose t is an odd integer. Then $\iota = \frac{t-1}{2}$ and $g(\mathbf{r}) = \oplus_{i=1}^{\iota} r_i r_{i+\iota} \oplus r_t$. For any $\mathbf{d} =$

$(\mathbf{d}', d_t) \in \mathbb{F}_2^{t-1} \times \mathbb{F}_2$, we have

$$\begin{aligned} W_g(\mathbf{d}) &= \sum_{\mathbf{r}\in\mathbb{F}_2^t} (-1)^{g(\mathbf{r})\oplus \mathbf{d}\cdot\mathbf{r}} = \sum_{(\mathbf{r}',r_t)\in\mathbb{F}_2^{t-1}\times\mathbb{F}_2} (-1)^{g(\mathbf{r}',r_t)\oplus\mathbf{d}'\cdot\mathbf{r}'\oplus d_t r_t} \\ &= \sum_{\mathbf{r}'\in\mathbb{F}_2^{t-1}} (-1)^{g_1(\mathbf{r}')\oplus\mathbf{d}'\cdot\mathbf{r}'} + \sum_{\mathbf{r}'\in\mathbb{F}_2^{t-1}} (-1)^{g_1(\mathbf{r}')\oplus 1\oplus\mathbf{d}'\cdot\mathbf{r}'\oplus d_t} \\ &= \left(1+(-1)^{d_t\oplus 1}\right) W_{g_1}(\mathbf{d}') \\ &= \begin{cases} 0, & \text{if } d_t = 0 \\ 2^{\frac{t+1}{2}}(-1)^{\bigoplus_{i=1}^{\iota} d_i d_{i+\iota}}, & \text{if } d_t = 1 \end{cases}. \end{aligned}$$

□

Thus, the maximum absolute Walsh-Hadamard spectrum value of $g \in \mathcal{B}_t$ is $2^{\frac{t}{2}+1}$ (and $2^{\frac{t+1}{2}}$) for t is even (odd, respectively).

Lemma 2. *Let $t \geq 5$ be an integer and $g \in \mathcal{B}_t$ be the function defined in Definition 1. Then for any $\mathbf{d} = (d_1, d_2, \cdots, d_t) \in \mathbb{F}_2^t$, if t is even, we have the autocorrelation value $C_g(\mathbf{d})$ of a function g at a point $\mathbf{d}$ as*

$$C_g(\mathbf{d}) \quad = \quad \begin{cases} 2^t(-1)^{d_{t-1}\oplus d_t}, & \text{if } (d_1, d_2 \ldots, d_{t-2}) = (0, 0, \ldots, 0) \\ 0, & \text{otherwise} \end{cases},$$

and if t is odd, we have

$$C_g(\mathbf{d}) \quad = \quad \begin{cases} 2^t(-1)^{d_t}, & \text{if } (d_1, d_2 \ldots, d_{t-1}) = (0, 0, \ldots, 0) \\ 0, & \text{otherwise,} \end{cases}.$$

Proof. Let $\iota = \lfloor \frac{t-1}{2} \rfloor$ and $\mathbf{b} = (d_{\iota+1}, \ldots, d_{2\iota}, d_1, \ldots, d_\iota)$ for $\mathbf{d} \in \mathbb{F}_2^t$. Suppose $\mathbf{r}' = (r_1, r_2, \ldots, r_{2\iota})$ for any $\mathbf{r} \in \mathbb{F}_2^t$. Let $t \geq 5$ be an even integer. Then $2\iota = t-2$ and

$$\begin{aligned} Cg(\mathbf{d}) &= \sum_{\mathbf{r}\in\mathbb{F}_2^t} (-1)^{g(\mathbf{r}\oplus\mathbf{d})\oplus g(\mathbf{r})} = \sum_{\mathbf{r}\in\mathbb{F}_2^t} (-1)^{\bigoplus_{i=1}^{\iota}\left((r_i\oplus d_i)(r_{i+\iota}\oplus d_{i+\iota})\oplus r_i r_{i+\iota}\right)\oplus d_{t-1}\oplus d_t} \\ &= \sum_{\mathbf{r}\in\mathbb{F}_2^t} (-1)^{\oplus_{i=1}^{\iota}(r_i d_{i+\iota}\oplus r_{i+\iota} d_i)\oplus\oplus_{i=1}^{\iota} d_i d_{i+\iota}\oplus d_{t-1}\oplus d_t} = (-1)^{g(\mathbf{d})} \sum_{\mathbf{r}\in\mathbb{F}_2^t} (-1)^{\mathbf{r}'\cdot\mathbf{b}} \\ &= \begin{cases} 2^t(-1)^{d_{t-1}\oplus d_t}, & \text{if } (d_1, d_2 \ldots, d_{t-2}) = (0, 0, \ldots, 0) \\ 0, & \text{if otherwise} \end{cases}. \end{aligned}$$

When $t \geq 5$ is an odd integer. Then we take $2\iota = t-1$, and follow the same line of argument to conclude the claim. □

Now, we define the two Boolean functions in $2t \geq 10$ variables, which are used to construct the sub-functions.

Definition 2. Let $k = 2t \geq 10$ be an even integer. We define two Boolean functions $p, q \in \mathcal{B}_k$ as follows: $p(\mathbf{z}, \mathbf{r}) = \bigoplus_{i=1}^t z_i r_i$, and

$$\begin{aligned} q(\mathbf{z}, \mathbf{r}) \quad &= \quad \begin{cases} g(\mathbf{r}), & \text{if } \mathbf{z} = \mathbf{0} \\ \bigoplus_{i=2}^{t} z_i r_{i-1} \oplus z_1 r_t \oplus z_2 r_t, & \text{otherwise} \end{cases} \\ &= \quad g(\mathbf{r}) \prod_{i=1}^{t} (z_i \oplus 1) \oplus \bigoplus_{i=2}^{t} z_i r_{i-1} \oplus z_1 r_t \oplus z_2 r_t, \end{aligned}$$

where $\mathbf{z} = (z_1, z_2, \ldots, z_t), \mathbf{r} = (r_1, r_2, \ldots, r_t) \in \mathbb{F}_2^t$ and $g \in \mathcal{B}_t$ is given in Definition 1.

Here, p is a quadratic MM function in $2t$ variables, and the algebraic degree of $q \in \mathcal{B}_{2t}$ is $t+2$. If we consider a zero function for $\mathbf{z} = \mathbf{0}$ instead of g, then q is also a quadratic bent function.

Lemma 3. *Let $t \geq 5$ be an arbitrary integer and $\phi(\mathbf{z}) = (z_2, z_3, \ldots, z_{t-1}, z_t, z_1 \oplus z_2)$, where $\mathbf{z} = (z_1, z_2, \ldots, z_t) \in \mathbb{F}_2^t$. Then both $\phi(\mathbf{z})$ and $\phi(\mathbf{z}) \oplus \mathbf{z}$ are permutations over $\mathbb{F}_2^t$. Moreover, the composition inverse of $\phi(\mathbf{z})$ is $\phi^{-1}(\mathbf{z}) = (z_1 \oplus z_t, z_1, z_2, \ldots, z_{t-1})$.*

Proof. It is clear that ϕ is a permutation over $\mathbb{F}_2^t$. Let $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^t$ such that $\phi(\mathbf{x}) \oplus \mathbf{x} = \phi(\mathbf{y}) \oplus \mathbf{y}$. Then

$$\begin{aligned}
& (x_1 \oplus x_2, \ldots, x_{t-1} \oplus x_t, x_1 \oplus x_2 \oplus x_t) = (y_1 \oplus y_2, \ldots, y_{t-1} \oplus y_t, y_1 \oplus y_2 \oplus y_t) \\
\Leftrightarrow\ & x_i \oplus x_{i+1} = y_i \oplus y_{i+1}, \text{ for all } 1 \leq i \leq t-1 \text{ and } x_1 \oplus x_2 \oplus x_t = y_1 \oplus y_2 \oplus y_t \\
\Leftrightarrow\ & x_i = y_i, \text{ for all } 1 \leq i \leq t.
\end{aligned}$$

Further, from the construction itself, the composition inverse functions of ϕ is indeed $\phi^{-1}(\mathbf{z}) = (z_1 \oplus z_t, z_1, \ldots, z_{t-1})$. □

Lemma 4. *Let $k = 2t \geq 10$ be an even integer and $p, q \in \mathcal{B}_k$ be the two functions defined in Definition 2. Then for any $\mathbf{c} = (c_1, c_2, \ldots, c_t), \mathbf{d} = (d_1, d_2, \ldots, d_t) \in \mathbb{F}_2^t$ we have*

$$W_p(\mathbf{c}, \mathbf{d}) = (-1)^{\mathbf{c}\cdot\mathbf{d}} \cdot 2^t,$$

$$W_q(\mathbf{c}, \mathbf{d}) = \begin{cases} 0, & \text{if } \mathbf{d} = \mathbf{0} \\ (-1)^{\mathbf{c}\cdot\mathbf{d}'} \cdot 2^t + W_g(\mathbf{d}), & \text{otherwise} \end{cases}, \text{and}$$

$$W_{p\oplus q}(\mathbf{c}, \mathbf{d}) = \begin{cases} 0, & \text{if } \mathbf{d} = \mathbf{0} \\ (-1)^{\mathbf{c}\cdot\mathbf{d}''} \cdot 2^t + W_g(\mathbf{d}), & \text{otherwise} \end{cases}.$$

$W_g(\mathbf{d})$ is given in Lemma 1, $\mathbf{d}'$ and $\mathbf{d}''$ are the composition inverse of $\phi(\mathbf{d})$ and $\phi(\mathbf{d}) \oplus \mathbf{d}$, respectively, where $\phi(\mathbf{d}) = (d_2, d_3, \ldots, d_{t-1}, d_t, d_1 \oplus d_2)$.

Proof. Since p is a bent function in $2t$ variables, for any $\mathbf{c}, \mathbf{d} \in \mathbb{F}_2^t$, we have $W_p(\mathbf{c}, \mathbf{d}) = 2^t(-1)^{\mathbf{c}\cdot\mathbf{d}}$. For any $\mathbf{c}, \mathbf{d} \in \mathbb{F}_2^t$,

$$\begin{aligned}
W_q(\mathbf{c}, \mathbf{d}) &= \sum_{\mathbf{z},\mathbf{r} \in \mathbb{F}_2^t} (-1)^{q(\mathbf{z},\mathbf{r}) \oplus \mathbf{c}\cdot\mathbf{z} \oplus \mathbf{d}\cdot\mathbf{r}} \\
&= \sum_{\mathbf{r} \in \mathbb{F}_2^t} (-1)^{g(\mathbf{r}) \oplus \mathbf{d}\cdot\mathbf{r}} + \sum_{\mathbf{z} \in \mathbb{F}_2^{t*}} \sum_{\mathbf{r} \in \mathbb{F}_2^t} (-1)^{\bigoplus_{i=2}^t z_i r_{i-1} \oplus z_1 r_t \oplus z_2 r_t \oplus \mathbf{c}\cdot\mathbf{z} \oplus \mathbf{d}\cdot\mathbf{r}} \\
&= W_g(\mathbf{d}) + \sum_{\mathbf{z} \in \mathbb{F}_2^{t*}} (-1)^{\mathbf{c}\cdot\mathbf{z}} \sum_{\mathbf{r} \in \mathbb{F}_2^t} (-1)^{\bigoplus_{i=2}^t z_i r_{i-1} \oplus z_1 r_t \oplus z_2 r_t \oplus \mathbf{d}\cdot\mathbf{r}} \\
&= W_g(\mathbf{d}) + \sum_{\mathbf{z} \in \mathbb{F}_2^{t*}} (-1)^{\mathbf{c}\cdot\mathbf{z}} \sum_{\mathbf{r} \in \mathbb{F}_2^t} (-1)^{(\mathbf{a}\oplus\mathbf{d})\cdot\mathbf{r}} = W_g(\mathbf{d}) + 2^t(-1)^{\mathbf{c}\cdot\mathbf{d}'},
\end{aligned}$$

where $\mathbf{a} = (z_2, z_3, \ldots, z_t, z_1 \oplus z_2)$ and $\mathbf{d}' = (d_1 \oplus d_t, d_1, \ldots, d_t)$. Since $W_g(\mathbf{0}) = 0$ and

$\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\bigoplus_{i=2}^t z_ir_{i-1}\oplus z_1r_t\oplus z_2r_t} = 0$ for $\mathbf{z}\neq\mathbf{0}$, we have $W_q(\mathbf{c},\mathbf{0})=0$. Also

$$\begin{aligned}
W_{p\oplus q}(\mathbf{c},\mathbf{d}) &= \sum_{\mathbf{z},\mathbf{r}\in\mathbb{F}_2^t}(-1)^{p(\mathbf{z},\mathbf{r})\oplus q(\mathbf{z},\mathbf{r})\oplus \mathbf{c}\cdot\mathbf{z}\oplus\mathbf{d}\cdot\mathbf{r}}\\
&= \sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{g(\mathbf{r})\oplus\mathbf{d}\cdot\mathbf{r}} + \sum_{\mathbf{z}\in\mathbb{F}_2^{t*}}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\mathbf{z}\cdot\mathbf{r}\oplus\bigoplus_{i=2}^t z_ir_{i-1}\oplus z_1r_t\oplus z_2r_t\oplus\mathbf{c}\cdot\mathbf{z}\oplus\mathbf{d}\cdot\mathbf{r}}\\
&= W_g(\mathbf{d}) + \sum_{\mathbf{z}\in\mathbb{F}_2^{t*}}(-1)^{\mathbf{c}\cdot\mathbf{z}}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\mathbf{z}\cdot\mathbf{r}\oplus\bigoplus_{i=2}^t z_ir_{i-1}\oplus z_1r_t\oplus z_2r_t\oplus\mathbf{d}\cdot\mathbf{r}}\\
&= W_g(\mathbf{d}) + \sum_{\mathbf{z}\in\mathbb{F}_2^{t*}}(-1)^{\mathbf{c}\cdot\mathbf{z}}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{(\mathbf{b}\oplus\mathbf{d})\cdot\mathbf{r}} = W_g(\mathbf{d}) + 2^t(-1)^{\mathbf{c}\cdot\mathbf{d}''},
\end{aligned}$$

where $\mathbf{b}=(z_1\oplus z_2,\dots,z_{t-1}\oplus z_t, z_1\oplus z_2\oplus z_t)$ and $\mathbf{d}''=(\bigoplus_{i=2}^t d_i, \bigoplus_{i=1}^t d_i, d_1\oplus\bigoplus_{i=3}^t d_i,\dots,d_1\oplus d_t)$. Since $W_g(\mathbf{0})=0$ and $\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\mathbf{z}\cdot\mathbf{r}\oplus\bigoplus_{i=2}^t z_ir_{i-1}\oplus z_1r_t\oplus z_2r_t}=0$ for $\mathbf{z}\neq\mathbf{0}$, we have $W_{p\oplus q}(\mathbf{c},\mathbf{0})=0$. □

It is clear that $|W_p(\mathbf{c},\mathbf{d})|=2^t$ for all $\mathbf{c},\mathbf{d}\in\mathbb{F}_2^t$, and

$$|W_q(\mathbf{c},\mathbf{d})|, |W_{p\oplus q}(\mathbf{c},\mathbf{d})| \le \begin{cases} 2^t+2^{\frac{t}{2}+1}, & \text{if } t \text{ is even}\\ 2^t+2^{\frac{t+1}{2}}, & \text{if } t \text{ is odd}\end{cases}.$$

Lemma 5. *Let $k=2t\ge 10$ be an even integer and $p,q\in\mathcal{B}_k$ be the two functions defined in Definition 2. Then for any $\mathbf{c}=(c_1,c_2,\dots,c_t),\mathbf{d}=(d_1,d_2,\dots,d_t)\in\mathbb{F}_2^t$ we have*

$$C_p(\mathbf{c},\mathbf{d}) = \begin{cases} 2^k, & \text{if } \mathbf{c}=\mathbf{d}=\mathbf{0}\\ 0, & \text{otherwise}\end{cases},$$

$$C_q(\mathbf{c},\mathbf{d}) = \begin{cases} 2^k, & \text{if } \mathbf{c}=\mathbf{d}=\mathbf{0}\\ C_g(\mathbf{d})-2^t, & \text{if } \mathbf{c}=\mathbf{0},\mathbf{d}\neq\mathbf{0}\\ 2(-1)^{\mathbf{c}'\cdot\mathbf{d}}W_g(\mathbf{c}'), & \text{if } \mathbf{c}\neq\mathbf{0}\end{cases},$$

where $\mathbf{c}'=(c_2,c_3,\dots,c_t,c_1\oplus c_2)$ for $\mathbf{c}\in\mathbb{F}_2^t$, and $W_g(\mathbf{c}')$ and $C_g(\mathbf{d})$ are given in Lemma 1 and Lemma 2, respectively.

Proof. Since p is a bent function in $2t$ variables, $C_p(\mathbf{0},\mathbf{0})=2^{2t}$ and $C_p(\mathbf{c},\mathbf{d})=0$ for all nonzero $\mathbf{c},\mathbf{d}\in\mathbb{F}_2^t$. we have derived the autocorrelation values of q in different cases.
Case (i): Let $\mathbf{c}=\mathbf{0}$. Then

$$q(\mathbf{z},\mathbf{r})\oplus q(\mathbf{z},\mathbf{r}\oplus\mathbf{d}) = \begin{cases} g(\mathbf{r})\oplus g(\mathbf{r}\oplus\mathbf{d}), & \text{if } \mathbf{z}=\mathbf{0}\\ \oplus_{i=2}^t z_id_{i-1}\oplus z_1d_t\oplus z_2d_t, & \text{otherwise}\end{cases}, \text{ and so,}$$

$$\begin{aligned}
C_q(\mathbf{0},\mathbf{d}) &= \sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{g(\mathbf{r})\oplus g(\mathbf{r}\oplus\mathbf{d})} + \sum_{\mathbf{z}\in\mathbb{F}_2^{t*}}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\oplus_{i=2}^t z_id_{i-1}\oplus z_1d_t\oplus z_2d_t}\\
&= C_g(\mathbf{d}) + 2^t\left(\sum_{\mathbf{z}\in\mathbb{F}_2^t}(-1)^{\mathbf{d}\cdot\mathbf{z}'}-1\right), \text{ where } \mathbf{z}'=(z_2,z_3,\dots,z_t,z_1\oplus z_2)\\
&= C_g(\mathbf{d}) + 2^t\sum_{\mathbf{z}\in\mathbb{F}_2^t}(-1)^{\mathbf{d}\cdot\mathbf{z}'}-2^t.
\end{aligned}$$

Here $(z_1,\dots,z_{t-1},z_t)\longmapsto(z_2,\dots,z_t,z_1\oplus z_2)$ is a permutation over $\mathbb{F}_2^t$. If $\mathbf{d}=\mathbf{0}$, then $C_q(\mathbf{0},\mathbf{0})=2^k$, and if $\mathbf{d}\neq\mathbf{0}$, then $C_q(\mathbf{0},\mathbf{d})=C_g(\mathbf{d})-2^t$.

Case (ii): Let $\mathbf{c} \neq \mathbf{0}$ and $\mathbf{d} = \mathbf{0}$. Suppose $\mathbf{c}' = (c_2, c_3, \ldots, c_t, c_1 \oplus c_2)$ for any nonzero $\mathbf{c} \in \mathbb{F}_2^t$. Then

$$\begin{aligned} C_q(\mathbf{c}, \mathbf{0}) &= \sum_{\mathbf{z},\mathbf{r}\in\mathbb{F}_2^t} (-1)^{q(\mathbf{z},\mathbf{r})\oplus q(\mathbf{z}\oplus\mathbf{c},\mathbf{r})} \\ &= 2\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{g(\mathbf{r})\oplus(\oplus_{i=2}^t c_i r_{i-1})\oplus c_1 r_t \oplus c_2 r_t} + \sum_{\mathbf{z}\in\mathbb{F}_2^t\setminus\{\mathbf{0},\mathbf{c}\}}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\oplus_{i=2}^t c_i r_{i-1}\oplus c_1 r_t\oplus c_2 r_t} \\ &= 2\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{g(\mathbf{r})\oplus\mathbf{c}'\cdot\mathbf{r}} + (2^t-2)\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\mathbf{c}'\cdot\mathbf{r}} \\ &= 2W_g(\mathbf{c}') + (2^t-2)\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\mathbf{c}'\cdot\mathbf{r}} = 2W_g(\mathbf{c}'), \end{aligned}$$

as $\mathbf{c}$ is nonzero. Case (iii): Let $\mathbf{c} \neq \mathbf{0}$ and $\mathbf{d} \neq \mathbf{0}$. Then we have

$$\begin{aligned} q(\mathbf{z},\mathbf{r}) \oplus q(\mathbf{z}\oplus\mathbf{c},\mathbf{r}\oplus\mathbf{d}) &= \begin{cases} g(\mathbf{r}) \oplus \oplus_{i=2}^t c_i r_{i-1} \oplus c_1 r_t \oplus c_2 r_t \oplus \\ \oplus_{i=2}^t c_i d_{i-1} \oplus c_1 d_t \oplus c_2 d_t, & \text{if } \mathbf{z} = \mathbf{0} \\ g(\mathbf{r}\oplus\mathbf{d}) \oplus \oplus_{i=2}^t c_i r_{i-1} \oplus c_1 r_t \oplus c_2 r_t, & \text{if } \mathbf{z} = \mathbf{c} \\ \oplus_{i=2}^t z_i d_{i-1} \oplus z_1 d_t \oplus z_2 d_t \oplus \\ \oplus_{i=2}^t c_i r_{i-1} \oplus c_1 r_t \oplus c_2 r_t \oplus \\ \oplus_{i=2}^t c_i d_{i-1} \oplus c_1 d_t \oplus c_2 d_t, & \text{if } \mathbf{z} \neq \mathbf{0}, \mathbf{c} \end{cases}, \\ &= \begin{cases} g(\mathbf{r}) \oplus \mathbf{c}'\cdot\mathbf{r} \oplus \mathbf{c}'\cdot\mathbf{d}, & \text{if } \mathbf{z} = \mathbf{0} \\ g(\mathbf{r}\oplus\mathbf{d}) \oplus \mathbf{c}'\cdot\mathbf{r}, & \text{if } \mathbf{z} = \mathbf{c} \\ \mathbf{z}'\cdot\mathbf{d} \oplus \mathbf{c}'\cdot\mathbf{r} \oplus \mathbf{c}'\cdot\mathbf{d}, & \text{if } \mathbf{z} \neq \mathbf{0}, \mathbf{c} \end{cases}, \end{aligned}$$

where $\mathbf{z}' = (z_2, z_3, \ldots, z_t, z_1 \oplus z_2)$ for $\mathbf{z} \in \mathbb{F}_2^n$, and so

$$\begin{aligned} C_q(\mathbf{c}, \mathbf{d}) &= \sum_{\mathbf{z},\mathbf{r}\in\mathbb{F}_2^t} (-1)^{q(\mathbf{z},\mathbf{r})\oplus q(\mathbf{z}\oplus\mathbf{c},\mathbf{r}\oplus\mathbf{d})} \\ &= \sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{g(\mathbf{r})\oplus\mathbf{c}'\cdot\mathbf{r}\oplus\mathbf{c}'\cdot\mathbf{d}} + \sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{g(\mathbf{r}\oplus\mathbf{d})\oplus\mathbf{c}'\cdot\mathbf{r}} \\ &\quad + \sum_{\mathbf{z}\in\mathbb{F}_2^t\setminus\{\mathbf{0},\mathbf{c}\}}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\mathbf{z}'\cdot\mathbf{d}\oplus\mathbf{c}'\cdot\mathbf{r}\oplus\mathbf{c}'\cdot\mathbf{d}} = 2(-1)^{\mathbf{c}'\cdot\mathbf{d}}W_g(\mathbf{c}'), \text{ as } \mathbf{c} \neq \mathbf{0}. \end{aligned}$$

□

Definition 3. Let $k = 2t \geq 10$ be an even integer. We define two functions $u, v \in \mathcal{B}_k$ as follows: $u(\mathbf{z},\mathbf{r}) = p(\mathbf{z},\mathbf{r})q(\mathbf{z},\mathbf{r}) = \bigoplus_{i=1}^t z_i r_i \bigoplus_{i=2}^{t} z_i r_{i-1} \oplus (z_1 r_t \oplus z_2 r_t) \bigoplus_{i=1}^t z_i r_i$, and

$$\begin{aligned} v(\mathbf{z},\mathbf{r}) &= \big(p(\mathbf{z},\mathbf{r}) \oplus 1\big)q(\mathbf{z},\mathbf{r}) = u(\mathbf{z},\mathbf{r}) \oplus q(\mathbf{z},\mathbf{r}) \\ &= g(\mathbf{r})\prod_{i=1}^t(z_i \oplus 1) \oplus \Big(\bigoplus_{i=1}^t z_i r_i \oplus 1\Big)\bigoplus_{i=2}^t z_i r_{i-1} \oplus (z_1 r_t \oplus z_2 r_t)\Big(\bigoplus_{i=1}^t z_i r_i \oplus 1\Big), \end{aligned}$$

where $\mathbf{z} = (z_1, z_2, \ldots, z_t), \mathbf{r} = (r_1, r_2, \ldots, r_t) \in \mathbb{F}_2^t$ and $g \in \mathcal{B}_t$ is given in Definition 1.

It is direct that the algebraic degree of the functions u and v in $2t \geq 10$ variables are 4 and $t+2$, respectively. Now, we derive the Walsh spectrum values of the functions u and v given in Definition 3 using the following known results.

Lemma 6 ([TKMM19]). *Let u_1, u_2 be two k-variable Boolean functions and define $u(\mathbf{x}) = u_1(\mathbf{x})u_2(\mathbf{x})$, where $\mathbf{x} = (x_1, x_2, \ldots, x_k) \in \mathbb{F}_2^k$. Then for any $\mathbf{b} \in \mathbb{F}_2^k$ we have $W_u(\mathbf{b}) =$*

$2^{k-1}\delta_{\mathbf{0}}(\mathbf{b}) + \frac{1}{2}\Big(W_{u_1}(\mathbf{b}) + W_{u_2}(\mathbf{b}) - W_{u_1\oplus u_2}(\mathbf{b})\Big)$, *where* $\delta_{\mathbf{0}}(\cdot)$ *is the Dirac (or Kronecker) symbol, which is defined by* $\delta_{\mathbf{0}}(\mathbf{b}) = 1$ *if* $\mathbf{b} = \mathbf{0}$ *and* $\delta_{\mathbf{0}}(\mathbf{b}) = 0$ *otherwise. For any* $\mathbf{d} \in \mathbb{F}_2^{k*}$, *we have*

$$\begin{aligned} C_u(\mathbf{d}) &= 2^{k-2} + \frac{1}{4}\Big(C_{u_1}(\mathbf{d}) + C_{u_2}(\mathbf{d}) + C_{u_1\oplus u_2}(\mathbf{d})\Big) + \frac{1}{2}\Big(W_{u_1}(\mathbf{0}) + W_{u_2}(\mathbf{0}) \\ &\quad -W_{u_1\oplus u_2}(\mathbf{0})\Big) + \frac{1}{2}\Big(C_{u_1,u_2}(\mathbf{d}) - C_{u_1,u_1\oplus u_2}(\mathbf{d}) - C_{u_2,u_1\oplus u_2}(\mathbf{d})\Big). \end{aligned}$$

Lemma 7. *Let* $k = 2t \geq 10$ *and* $u, v \in \mathcal{B}_k$ *be the functions defined by Definition 3. For any* $\mathbf{c}, \mathbf{d} \in \mathbb{F}_2^t$, *we have*

$$W_u(\mathbf{c},\mathbf{d}) = \begin{cases} 2^{k-1} + 2^{t-1}, & \text{if } \mathbf{c} = \mathbf{d} = \mathbf{0} \\ 2^{t-1}, & \text{if } \mathbf{c} \neq \mathbf{0}, \mathbf{d} = \mathbf{0} \\ 2^{t-1}((-1)^{\mathbf{c}\cdot\mathbf{d}} + (-1)^{\mathbf{c}\cdot\mathbf{d}'} - (-1)^{\mathbf{c}\cdot\mathbf{d}''}), & \text{if } \textit{otherwise} \end{cases}, \textit{ and}$$

$$W_v(\mathbf{c},\mathbf{d}) = \begin{cases} 2^{k-1} - 2^{t-1}, & \text{if } \mathbf{c} = \mathbf{d} = \mathbf{0} \\ -2^{t-1}, & \text{if } \mathbf{c} \neq \mathbf{0}, \mathbf{d} = \mathbf{0} \\ W_g(\mathbf{d}) - 2^{t-1}((-1)^{\mathbf{c}\cdot\mathbf{d}} - (-1)^{\mathbf{c}\cdot\mathbf{d}'} - (-1)^{\mathbf{c}\cdot\mathbf{d}''}), & \text{if } \textit{otherwise} \end{cases},$$

where $\mathbf{d}' = (d_1 \oplus d_t, d_1, \ldots, d_{t-1})$ *and* $\mathbf{d}'' = (\bigoplus_{i=2}^t d_i, \bigoplus_{i=1}^t d_i, d_1 \oplus \bigoplus_{i=3}^t d_i, \ldots, d_1 \oplus d_t)$.

Proof. From [TKMM19, Theorem 20] and Lemma 4, we have for any $\mathbf{c}, \mathbf{d} \in \mathbb{F}_2^t$

$$\begin{aligned} W_u(\mathbf{c},\mathbf{d}) &= 2^{k-1}\delta_{\mathbf{0}}(\mathbf{c},\mathbf{d}) + \frac{1}{2}\big(W_p(\mathbf{c},\mathbf{d}) + W_q(\mathbf{c},\mathbf{d}) - W_{p\oplus q}(\mathbf{c},\mathbf{d})\big) \\ &= 2^{k-1}\delta_{\mathbf{0}}(\mathbf{c},\mathbf{d}) + 2^{t-1}(-1)^{\mathbf{c}\cdot\mathbf{d}} + \begin{cases} 0, & \text{if } \mathbf{d} = \mathbf{0} \\ 2^{t-1}(-1)^{\mathbf{c}\cdot\mathbf{d}'} + \frac{W_g(\mathbf{d})}{2}, & \text{otherwise} \end{cases} \\ &\quad - \begin{cases} 0, & \text{if } \mathbf{d} = \mathbf{0} \\ 2^{t-1}(-1)^{\mathbf{c}\cdot\mathbf{d}''} + \frac{W_g(\mathbf{d})}{2}, & \text{otherwise} \end{cases} \\ &= \begin{cases} 2^{k-1} + 2^{t-1}, & \text{if } \mathbf{c} = \mathbf{d} = \mathbf{0} \\ 2^{t-1}, & \text{if } \mathbf{c} \neq \mathbf{0}, \mathbf{d} = \mathbf{0} \\ 2^{t-1}((-1)^{\mathbf{c}\cdot\mathbf{d}} + (-1)^{\mathbf{c}\cdot\mathbf{d}'} - (-1)^{\mathbf{c}\cdot\mathbf{d}''}), & \text{if otherwise} \end{cases}. \end{aligned}$$

where $\mathbf{d}' = (d_1 \oplus d_t, d_1, \ldots, d_t)$ and $\mathbf{d}'' = (\bigoplus_{i=2}^t d_i, \bigoplus_{i=1}^t d_i, d_1 \oplus \bigoplus_{i=3}^t d_i, \ldots, d_1 \oplus d_t)$. Since $W_{p\oplus 1}(\mathbf{c},\mathbf{d}) = -W_p(\mathbf{c},\mathbf{d})$ for all $\mathbf{c}, \mathbf{d} \in \mathbb{F}_2^t$, we have

$$\begin{aligned} W_v(\mathbf{c},\mathbf{d}) &= 2^{k-1}\delta_{\mathbf{0}}(\mathbf{c},\mathbf{d}) + \frac{1}{2}\big(W_{p\oplus 1}(\mathbf{c},\mathbf{d}) + W_q(\mathbf{c},\mathbf{d}) - W_{p\oplus q\oplus 1}(\mathbf{c},\mathbf{d})\big) \\ &= 2^{k-1}\delta_{\mathbf{0}}(\mathbf{c},\mathbf{d}) + \frac{1}{2}\big(- W_p(\mathbf{c},\mathbf{d}) + W_q(\mathbf{c},\mathbf{d}) + W_{p\oplus q}(\mathbf{c},\mathbf{d})\big) \\ &= \begin{cases} 2^{k-1} - 2^{t-1}, & \text{if } \mathbf{c} = \mathbf{d} = \mathbf{0} \\ -2^{t-1}, & \text{if } \mathbf{c} \neq \mathbf{0}, \mathbf{d} = \mathbf{0} \\ W_g(\mathbf{d}) - 2^{t-1}((-1)^{\mathbf{c}\cdot\mathbf{d}} - (-1)^{\mathbf{c}\cdot\mathbf{d}'} - (-1)^{\mathbf{c}\cdot\mathbf{d}''}), & \text{if otherwise} \end{cases}. \end{aligned}$$

□

From the above results, it is clear that the maximum absolute Walsh spectrum values of the functions u and v in $k = 2t \geq 10$ variables are $2^{k-1}+2^{t-1}$ and $2^{k-1}-2^{t-1}$, respectively.

Lemma 8. *Let* $k = 2t \geq 10$ *and* $u, v \in \mathcal{B}_k$ *be the functions defined by Definition 3. For any* $\mathbf{c}, \mathbf{d} \in \mathbb{F}_2^t$, *we have*

$$\begin{aligned} 2^{k-2} - 2^t \leq& C_u(\mathbf{c},\mathbf{d}) \leq 2^{k-2} + 2^{t+2}, \textit{ and} \\ 2^{k-2} - 3\cdot 2^{t+1} \leq& C_v(\mathbf{c},\mathbf{d}) \leq 2^{k-2} + 3\cdot 2^{t+1}. \end{aligned}$$

Proof. We first consider the autocorrelation spectrum of the Boolean function u. Let us define $q'(\mathbf{z},\mathbf{r}) = \bigoplus_{i=2}^{t} z_i r_{i-1} \oplus z_1 r_t \oplus z_2 r_t = \phi(\mathbf{z})\cdot\mathbf{r}$ which is a quadratic bent function, where $\phi(\mathbf{z}) = (z_2, z_3, \ldots, z_{t-1}, z_t, z_1 \oplus r_2)$ is a permutation over $\mathbb{F}_2^t$. Then we have $u(\mathbf{z},\mathbf{r}) = p(\mathbf{z},\mathbf{r})q(\mathbf{z},\mathbf{r}) = p(\mathbf{z},\mathbf{r})q'(\mathbf{z},\mathbf{r})$ since $p(\mathbf{z},\mathbf{r}) = 0$ with $\mathbf{z} = \mathbf{0}$. Obviously, we have $C_u(\mathbf{c},\mathbf{d}) = 2^k$ if $(\mathbf{c},\mathbf{d}) = (\mathbf{0},\mathbf{0})$. In what follows, we consider the values of $C_u(\mathbf{c},\mathbf{d})$ with $(\mathbf{c},\mathbf{d}) \in \mathbb{F}_2^k \setminus \{\mathbf{0},\mathbf{0}\}$ and our strategy is based on Lemma 4.6. First, we can easily see that $W_p(\mathbf{0},\mathbf{0}) = W_{q'}(\mathbf{0},\mathbf{0}) = W_{p\oplus q'}(\mathbf{0},\mathbf{0}) = 2^t$. In addition, for any $(\mathbf{c},\mathbf{d}) \in \mathbb{F}_2^k \setminus \{\mathbf{0},\mathbf{0}\}$, we have $C_p(\mathbf{c},\mathbf{d}) = C_{q'}(\mathbf{c},\mathbf{d}) = 0$ and $C_{p\oplus q'}(\mathbf{c},\mathbf{d}) = 0$ since $(p \oplus q')(\mathbf{z},\mathbf{r})$ is also a bent function due to $\mathbf{z} \oplus \phi(\mathbf{z})$ is a permutation on $\mathbb{F}_2^t$ by Lemma 3. Furthermore, we have

$$\begin{aligned}
C_{p,q'}(\mathbf{c},\mathbf{d}) &= \sum_{\mathbf{z}\in\mathbb{F}_2^t}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\mathbf{z}\cdot\mathbf{r}\oplus(\phi(\mathbf{z}\oplus\mathbf{c})\cdot(\mathbf{r}\oplus\mathbf{d}))} = \sum_{\mathbf{z}\in\mathbb{F}_2^t}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{\mathbf{z}\cdot\mathbf{r}\oplus(\phi(\mathbf{z})\oplus\phi(\mathbf{c}))\cdot(\mathbf{r}\oplus\mathbf{d}))} \\
&= \sum_{\mathbf{z}\in\mathbb{F}_2^t}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{(\mathbf{z}\oplus\phi(\mathbf{z}))\cdot\mathbf{r}\oplus\phi(\mathbf{z})\cdot\mathbf{d}\oplus\phi(\mathbf{c})\cdot\mathbf{r}\oplus\phi(\mathbf{c})\cdot\mathbf{d}} \\
&= \sum_{\mathbf{z}\in\mathbb{F}_2^t}\sum_{\mathbf{r}\in\mathbb{F}_2^t}(-1)^{(\phi^{-1}(\mathbf{z})\oplus\mathbf{z})\cdot\mathbf{r}\oplus\mathbf{z}\cdot\mathbf{d}\oplus\phi(\mathbf{c})\cdot\mathbf{r}\oplus\phi(\mathbf{c})\cdot\mathbf{d}} \\
&= (-1)^{\phi(\mathbf{c})\cdot\mathbf{d}}W_{\phi'(\mathbf{z})\cdot\mathbf{r}}(\mathbf{d},\phi(\mathbf{c})) \in \{2^t, -2^t\},
\end{aligned}$$

where $\phi'(\mathbf{z}) = \phi^{-1}(\mathbf{z}) \oplus \mathbf{z}$ is a permutation over $\mathbb{F}_2^t$. Similarly, we have $C_{p,p\oplus q'}(\mathbf{c},\mathbf{d}) \in \{2^t, -2^t\}$ and $C_{q',p\oplus q'}(\mathbf{c},\mathbf{d}) \in \{2^t, -2^t\}$. Therefore, for any $(\mathbf{c},\mathbf{d}) \in \mathbb{F}_2^k \setminus \{\mathbf{0},\mathbf{0}\}$ we have

$$2^{k-2} - 2^t \le C_{pq}(\mathbf{c},\mathbf{d}) = C_{pq'}(\mathbf{c},\mathbf{d}) \le 2^{k-2} + 2^{t+2}.$$

We now consider the autocorrelation spectrum of the Boolean function v. Similar to the case of u. We can get that

$$2^{k-2} - 2^{t+1} \le C_{(p\oplus 1)q'}(\mathbf{c},\mathbf{d}) \le 2^{k-2} + 2^{t+1}.$$

Note that the support of $v = (p\oplus 1)q$ equals $\mathrm{supp}((p\oplus 1)q') \cup \mathrm{supp}(g)$ and $|\mathrm{supp}(g)| = 2^{t-1}$. Then we have

$$2^{k-2} - 3\cdot 2^{t+1} \le C_{(p\oplus 1)q}(\mathbf{c},\mathbf{d}) \le 2^{k-2} + 3\cdot 2^{t+1}.$$

This completes the proof. □

3.2 Main construction of ultra-lightweight Boolean functions

In this section, we construct a class of balanced Boolean functions by modifying a simple MM function using two sub-functions defined in Definition 3.

Construction 1. *Let $n = 2k = 4t$ be an even integer not less than 20. We construct an n-variable Boolean function f over $\mathbb{F}_2^k \times \mathbb{F}_2^k$ as follows*

$$f(\mathbf{x},\mathbf{y}) = \begin{cases} u(\mathbf{y}), & \text{if } \mathbf{x} = \mathbf{1}, \mathbf{y} \in \mathbb{F}_2^k \setminus \{\mathbf{1}\} \\ v(\mathbf{x}), & \text{if } \mathbf{y} = \mathbf{x} \in \mathbb{F}_2^k \\ (\mathbf{x}\oplus\mathbf{y})\cdot\mathbf{y}, & \text{otherwise} \end{cases},$$

where u and v are the two Boolean functions over $\mathbb{F}_2^k$ defined in Definition 3.

Since the outputs of MM function $(\mathbf{x}\oplus\mathbf{y})\cdot\mathbf{y} = \mathbf{x}\cdot\mathbf{y} \oplus \bigoplus_{i=1}^k y_i$ is modified by a sub-function u when $\mathbf{x} = \mathbf{1}$ and $\mathbf{y} \neq \mathbf{1}$, and a sub-function v when $\mathbf{x} = \mathbf{y}$, where $\mathbf{x},\mathbf{y} \in \mathbb{F}_2^k$. The function f can be written as

$$\begin{aligned}
f(\mathbf{x},\mathbf{y}) = {} & (x_1x_2\cdots x_k(y_1y_2\cdots y_k \oplus 1)\oplus 1)(\mathbf{x}\oplus\mathbf{y})\cdot\mathbf{y} \oplus x_1x_2\cdots x_k(y_1y_2\cdots y_k \\
& \oplus 1)u(\mathbf{y}) \oplus (x_1\oplus y_1\oplus 1)(x_2\oplus y_2\oplus 1)\cdots(x_k\oplus y_k\oplus 1)v(\mathbf{x})
\end{aligned}$$

3.3 Cryptographic Properties of the Construction 1

We analyze the Boolean function generated in Construction 1 with respect to three key properties: balancedness, nonlinearity, and absolute indicator, each considered separately.

3.3.1 Balancedness

We first prove that the Boolean function generated in Construction 1 is balanced. We will use the following lemma to prove our claim.

Lemma 9. *Let $n = 2k \geq 20$ and $f \in \mathcal{B}_n$ be a Boolean function generated by Construction 1. Then for any $(\mathbf{a}, \mathbf{b}) \in \mathbb{F}_2^k \times \mathbb{F}_2^k$ we have*

$$W_f(\mathbf{a}, \mathbf{b}) = \begin{cases} W_u(\mathbf{b}) + W_v(\mathbf{b}) + 2^k, & \text{if } (\mathbf{a}, \mathbf{b}) \in \{\mathbf{0}\} \times \mathbb{F}_2^{k*} \\ W_u(\mathbf{0})(-1)^{\mathbf{1}\cdot\mathbf{a}} + W_v(\mathbf{a}), & \text{if } (\mathbf{a}, \mathbf{b}) \in \mathbb{F}_2^{k*} \times \{\mathbf{0}\} \\ W_u(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{a}} + W_v(\mathbf{a} \oplus \mathbf{b}) & \\ +2^k(-1)^{\mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})}, & \text{if } (\mathbf{a}, \mathbf{b}) \in \mathbb{F}_2^{k*} \times \mathbb{F}_2^{k*} \setminus U \\ W_u(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{b}} + W_v(\mathbf{0}) & \\ -2^k \delta_{\mathbf{0}}(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{b}}, & \text{if } (\mathbf{a}, \mathbf{b}) \in U \end{cases},$$

where $U = \{(\mathbf{c}, \mathbf{c}) : \mathbf{c} \in \mathbb{F}_2^k\} \subseteq \mathbb{F}_2^k \times \mathbb{F}_2^k$ and $\delta_{\mathbf{0}}(\cdot)$ is the Dirac (or Kronecker) symbol which is defined by $\delta_{\mathbf{0}}(\mathbf{b}) = 1$ if $\mathbf{b} = \mathbf{0}$ and $\delta_{\mathbf{0}}(\mathbf{b}) = 0$ otherwise.

Proof. We can easily get that $\sum_{\mathbf{x}\in\mathbb{F}_2^k}(-1)^{\mathbf{c}\cdot\mathbf{x}\oplus\mathbf{d}\cdot\mathbf{y}}$ equals 0 if $\mathbf{c} \in \mathbb{F}_2^{k*}$ and equals 2^k otherwise, where $\mathbf{d}$ and $\mathbf{y}$ are arbitrary vectors in $\mathbb{F}_2^k$. To proceed to calculate Walsh Hadamard spectrum of the constructed function, we partition the total sum into three disjoint components based on the structure of the input pairs $(\mathbf{x}, \mathbf{y})$, allowing us to isolate contributions from $(\mathbf{1}, \mathbf{y})$, the special set U, and the remaining pairs.

$$\begin{aligned} W_f(\mathbf{a}, \mathbf{b}) &= \sum_{(\mathbf{x},\mathbf{y})\in\mathbb{F}_2^k\times\mathbb{F}_2^k} (-1)^{f(\mathbf{x},\mathbf{y})\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}} \\ &= \sum_{(\mathbf{x},\mathbf{y})\in\{\mathbf{1}\}\times\mathbb{F}_2^k\setminus\{\mathbf{1}\}} (-1)^{f(\mathbf{x},\mathbf{y})\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}} + \sum_{(\mathbf{x},\mathbf{y})\in U} (-1)^{f(\mathbf{x},\mathbf{y})\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}} \\ &\quad + \sum_{(\mathbf{x},\mathbf{y})\in\{(\mathbf{x},\mathbf{y}):\mathbf{x}\in\mathbb{F}_2^k\setminus\{\mathbf{1}\},\mathbf{y}\in\mathbb{F}_2^k,\mathbf{y}\neq\mathbf{x}\}} (-1)^{f(\mathbf{x},\mathbf{y})\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}} \\ &= \sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{1}\}} (-1)^{u(\mathbf{y})\oplus\mathbf{1}\cdot\mathbf{a}\oplus\mathbf{b}\cdot\mathbf{y}} + \sum_{\mathbf{x}\in\mathbb{F}_2^k} (-1)^{v(\mathbf{x})\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{x}} \\ &\quad + \sum_{(\mathbf{x},\mathbf{y})\in\{(\mathbf{x},\mathbf{y}):\mathbf{x}\in\mathbb{F}_2^k\setminus\{\mathbf{1}\},\mathbf{y}\in\mathbb{F}_2^k,\mathbf{y}\neq\mathbf{x}\}} (-1)^{\mathbf{x}\cdot\mathbf{y}\oplus\mathbf{1}\cdot\mathbf{y}\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}}. \end{aligned}$$

Note that

$$\begin{aligned} \sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{1}\}} (-1)^{u(\mathbf{y})\oplus\mathbf{1}\cdot\mathbf{a}\oplus\mathbf{b}\cdot\mathbf{y}} &= W_u(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{a}} - (-1)^{u(\mathbf{1})\oplus\mathbf{1}\cdot(\mathbf{a}\oplus\mathbf{b})} \\ &= W_u(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{a}} - (-1)^{\mathbf{1}\cdot(\mathbf{a}\oplus\mathbf{b})} \text{ (since } u(\mathbf{1}) = 0). \end{aligned}$$

It is also easy to check that $\sum_{\mathbf{x}\in\mathbb{F}_2^k}(-1)^{v(\mathbf{x})\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{x}} = W_v(\mathbf{a} \oplus \mathbf{b})$.

We now simplify the third term of the sum by expressing it as the difference between the double sum over $\mathbb{F}_2^k \times \mathbb{F}_2^k$ and the special sets already accounted for. Specifically, we subtract out the contributions from $(\mathbf{1}, \mathbf{y})$ and the special set $U = \{(\mathbf{x}, \mathbf{x})\}$. This decomposition relies on the use of the orthogonality of characters and delta functions to simplify the expression as follows.

$$
\begin{aligned}
& \sum_{(\mathbf{x},\mathbf{y})\in\{(\mathbf{x},\mathbf{y}):\mathbf{x}\in\mathbb{F}_2^k\setminus\{\mathbf{1}\},\mathbf{y}\in\mathbb{F}_2^k,\mathbf{y}\neq\mathbf{x}\}} (-1)^{\mathbf{x}\cdot\mathbf{y}\oplus\mathbf{1}\cdot\mathbf{y}\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}} \\
= & \sum_{(\mathbf{x},\mathbf{y})\in\mathbb{F}_2^k\times\mathbb{F}_2^k\setminus U} (-1)^{\mathbf{x}\cdot\mathbf{y}\oplus\mathbf{1}\cdot\mathbf{y}\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}} - \sum_{(x,y)\in\{\mathbf{1}\}\times\mathbb{F}_2^k\setminus\{\mathbf{1}\}} (-1)^{\mathbf{x}\cdot\mathbf{y}\oplus\mathbf{1}\cdot\mathbf{y}\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}} \\
= & \left[\sum_{(\mathbf{x},\mathbf{y})\in\mathbb{F}_2^k\times\mathbb{F}_2^k} (-1)^{\mathbf{x}\cdot\mathbf{y}\oplus\mathbf{1}\cdot\mathbf{y}\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{y}} - \sum_{x\in\mathbb{F}_2^k} (-1)^{\mathbf{x}\cdot\mathbf{x}\oplus\mathbf{1}\cdot\mathbf{x}\oplus\mathbf{a}\cdot\mathbf{x}\oplus\mathbf{b}\cdot\mathbf{x}}\right] \\
& - \left[\sum_{y\in\mathbb{F}_2^k} (-1)^{\mathbf{1}_k\cdot\mathbf{y}\oplus\mathbf{1}\cdot\mathbf{y}\oplus\mathbf{a}\cdot\mathbf{1}\oplus\mathbf{b}\cdot\mathbf{y}} - (-1)^{\mathbf{1}\cdot(\mathbf{a}\oplus\mathbf{b})}\right] \\
= & \left[\sum_{x\in\mathbb{F}_2^k} (-1)^{\mathbf{a}\cdot\mathbf{x}} \sum_{y\in\mathbb{F}_2^k} (-1)^{(\mathbf{x}\oplus\mathbf{1}\oplus\mathbf{b})\cdot\mathbf{y}} - \sum_{x\in\mathbb{F}_2^k} (-1)^{(\mathbf{a}\oplus\mathbf{b})\cdot\mathbf{x}}\right] \\
& - \left[(-1)^{\mathbf{1}\cdot\mathbf{a}} \sum_{y\in\mathbb{F}_2^k} (-1)^{\mathbf{b}\cdot\mathbf{y}} - (-1)^{\mathbf{1}\cdot(\mathbf{a}\oplus\mathbf{b})}\right] \\
= & \ 2^k(-1)^{\mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})} - 2^k\delta_{\mathbf{0}}(\mathbf{a}\oplus\mathbf{b}) - 2^k\delta_{\mathbf{0}}(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{a}} + (-1)^{\mathbf{1}\cdot(\mathbf{a}\oplus\mathbf{b})}.
\end{aligned}
$$

Then we have

$$
\begin{aligned}
W_f(\mathbf{a},\mathbf{b}) &= W_u(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{a}} + W_v(\mathbf{a}\oplus\mathbf{b}) + 2^k(-1)^{\mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})} - 2^k\delta_{\mathbf{0}}(\mathbf{a}\oplus\mathbf{b}) \\
& \quad -2^k\delta_{\mathbf{0}}(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{a}} \\
&= \begin{cases}
W_u(\mathbf{b}) + W_v(\mathbf{b}) + 2^k, & \text{if } (\mathbf{a},\mathbf{b}) \in \{\mathbf{0}\}\times\mathbb{F}_2^{k*} \\
W_u(\mathbf{0})(-1)^{\mathbf{1}\cdot\mathbf{a}} + W_v(\mathbf{a}), & \text{if } (\mathbf{a},\mathbf{b}) \in \mathbb{F}_2^{k*}\times\{\mathbf{0}\} \\
W_u(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{a}} + W_v(\mathbf{a}\oplus\mathbf{b}) & \\
+2^k(-1)^{\mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})}, & \text{if } (\mathbf{a},\mathbf{b}) \in \mathbb{F}_2^{k*}\times\mathbb{F}_2^{k*}\setminus U \\
W_u(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{b}} + W_v(\mathbf{0}) & \\
-2^k\delta_{\mathbf{0}}(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{b}}, & \text{if } (\mathbf{a},\mathbf{b}) \in U
\end{cases}.
\end{aligned}
$$

□

Theorem 1. *Let $n = 2k = 4t$ be an even integer not less than 20 and t be an even integer. Then the n-variable Boolean function $f \in \mathcal{B}_n$ generated by Construction 1 is balanced.*

Proof. It follows from Lemma 9 that $W_f(\mathbf{0}) = W_u(\mathbf{0}) + W_v(\mathbf{0}) - 2^k = 0$, so we can conclude that the n-variable Boolean function $f \in \mathcal{B}_n$ generated by Construction 1 is balanced. □

3.3.2 Nonlinearity

In the following theorem, we derive the nonlinearity of balanced Boolean functions generated in Construction 1.

Theorem 2. *Let $n = 2k = 4t$ be an even integer not less than 20 and t be an even integer. The nonlinearity of n-variable Boolean function $f \in \mathcal{B}_n$ generated by Construction 1 is*

$$nl(f) = 2^{2k-1} - 2^{k-1} - 3\cdot 2^{t-1} - 2^{\frac{t}{2}}.$$

Further, if t is odd, the nonlinearity of f is

$$2^{2k-1} - 2^{k-1} - 3\cdot 2^{t-1} - 2^{\frac{t-1}{2}} \le nl(f) \le 2^{2k-1} - 2^{k-1} - 3\cdot 2^{t-1}.$$

Proof. Let $\mathbf{a} = \mathbf{0}$ and $\mathbf{b} \neq \mathbf{0} \in \mathbb{F}_2^k$. Suppose $\mathbf{b} = (\mathbf{c}, \mathbf{d}) \neq (\mathbf{0}, \mathbf{0}) \in \mathbb{F}_2^t \times \mathbb{F}_2^t$. If $\mathbf{d} = \mathbf{0}$, then $\mathbf{c} \neq \mathbf{0}$, and $W_u(\mathbf{c}, \mathbf{0}) + W_v(\mathbf{c}, \mathbf{0}) + 2^k = 2^k$. If $\mathbf{d} \neq \mathbf{0}$, then $W_u(\mathbf{c}, \mathbf{d}) + W_v(\mathbf{c}, \mathbf{d}) + 2^k = 2^k + W_g(\mathbf{d}) + 2^t(-1)^{\mathbf{c}\cdot\mathbf{d}'}$, where the value of $W_g(\mathbf{d})$ is given in Lemma 1. Thus,

$$\max_{\mathbf{c},\mathbf{d} \in \mathbb{F}_2^t} |W_f(\mathbf{0}, \mathbf{b})| = \begin{cases} 2^k + 2^t + 2^{\frac{t+1}{2}}, & \text{if } t \text{ is odd} \\ 2^k + 2^t + 2^{\frac{t}{2}+1}, & \text{if } t \text{ is even} \end{cases}.$$

Let $\mathbf{a} \neq \mathbf{0} \in \mathbb{F}_2^k$ and $\mathbf{b} = \mathbf{0}$. Suppose $\mathbf{a} = (\mathbf{c}, \mathbf{d}) \neq (\mathbf{0}, \mathbf{0}) \in \mathbb{F}_2^t \times \mathbb{F}_2^t$. Then $W_f(\mathbf{a}, \mathbf{0}) = W_u(\mathbf{0}, \mathbf{0})(-1)^{wt(\mathbf{c},\mathbf{d})} + W_v(\mathbf{c}, \mathbf{d})$, and so

$$\max_{\mathbf{c},\mathbf{d} \in \mathbb{F}_2^t} |W_f(\mathbf{a}, \mathbf{0})| = \begin{cases} 2^{k-1} + 2^{t+1} + 2^{\frac{t+1}{2}}, & \text{if } t \text{ is odd} \\ 2^{k-1} + 2^{t+1} + 2^{\frac{t}{2}+1}, & \text{if } t \text{ is even} \end{cases}.$$

Let $\mathbf{a} = \mathbf{b} = (\mathbf{c}, \mathbf{d}) \in \mathbb{F}_2^k$, where $\mathbf{c}, \mathbf{d} \in \mathbb{F}_2^t$. Then $W_f(\mathbf{a}, \mathbf{a}) = W_u(\mathbf{c}, \mathbf{d})(-1)^{wt(\mathbf{c},\mathbf{d})}$ $+W_v(\mathbf{0}, \mathbf{0}) - 2^k(-1)^{wt(\mathbf{c},\mathbf{d})}\delta_{(\mathbf{0},\mathbf{0})}(\mathbf{c}, \mathbf{d})$, and so $\max_{\mathbf{a} \in \mathbb{F}_2^k} |W_f(\mathbf{a}, \mathbf{a})| = 2^{k-1} + 2^t$. Let $\mathbf{a} \neq \mathbf{0}, \mathbf{b} \neq \mathbf{0} \in \mathbb{F}_2^k$ and $\mathbf{a} = \mathbf{b}$. Suppose $\mathbf{a} = (\mathbf{c}, \mathbf{d})$ and $\mathbf{b} = (\mathbf{e}, \mathbf{h}) \in \mathbb{F}_2^t \times \mathbb{F}_2^t$. Since $(\mathbf{d} \oplus \mathbf{h})' = \mathbf{d}' \oplus \mathbf{h}'$ and $(\mathbf{d} \oplus \mathbf{h})'' = \mathbf{d}'' \oplus \mathbf{h}''$. If t be an even integer, then there exist values of $\mathbf{a}, \mathbf{b}$ such that $wt(\mathbf{a})$ is even, $\mathbf{a} \cdot \mathbf{b} = 0$, $\mathbf{c} \cdot \mathbf{d} = 0$, $\mathbf{c} \cdot \mathbf{d}' = 0$, $\mathbf{c} \cdot \mathbf{d}'' = 1$, $(\mathbf{c} \oplus \mathbf{e}) \cdot (\mathbf{d} \oplus \mathbf{h}) = 1$, $(\mathbf{c} \oplus \mathbf{e}) \cdot (\mathbf{d}' \oplus \mathbf{h}') = 0$, $(\mathbf{c} \oplus \mathbf{e}) \cdot (\mathbf{d}'' \oplus \mathbf{h}'') = 0$ and $(d_{t-1} \oplus h_{t-1}, d_t \oplus h_t) = (1, 1)$. Then $\max_{\mathbf{a},\mathbf{b} \in \mathbb{F}_2^k} |W_f(\mathbf{a}, \mathbf{b})| = 2^k + 3 \cdot 2^t + 2^{\frac{t}{2}+1}$. Thus, the nonlinearity of f is $nl(f) = 2^{2k-1} - 2^{k-1} - 3 \cdot 2^{t-1} - 2^{\frac{t}{2}}$.

Let t be an odd integer. Then the maximum absolute value of $W_f(\mathbf{a}, \mathbf{b})$ satisfies the inequality $2^k + 3 \cdot 2^t \leq \max_{\mathbf{a} \neq \mathbf{b} \in \mathbb{F}_2^k \setminus \{\mathbf{0}\}} |W_f(\mathbf{a}, \mathbf{b})| \leq 2^k + 3 \cdot 2^t + 2^{\frac{t+1}{2}}$. Thus, the nonlinearity of f is $2^{2k-1} - 2^{k-1} - 3 \cdot 2^{t-1} - 2^{\frac{t-1}{2}} \leq nl(f) \leq 2^{2k-1} - 2^{k-1} - 3 \cdot 2^{t-1}$. □

3.3.3 Absolute Indicator

Here we compute the absolute indicator value of balanced Boolean function defined in Construction 1. For that, we first derive the following lemma.

Lemma 10. *Let $n = 2k = 4t \geq 8$ and $f \in \mathcal{B}_n$ be a Boolean function generated by Construction 1. Then for any $(\mathbf{a}, \mathbf{b}) \in \mathbb{F}_2^k \times \mathbb{F}_2^k$, we have*

$$C_f(\mathbf{a}, \mathbf{b}) = \begin{cases} 2^n, & \text{if } (\mathbf{a}, \mathbf{b}) = (\mathbf{0}, \mathbf{0}) \\ C_u(\mathbf{b}) + 2W_v(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{b}} & \\ -2(-1)^{u(\mathbf{1}) \oplus u(\mathbf{1} \oplus \mathbf{b})} - 2(-1)^{v(\mathbf{1})} & \\ +2(-1)^{v(\mathbf{1}) \oplus u(\mathbf{1} \oplus b)} - 2^k + 2, & \text{if } (\mathbf{a}, \mathbf{b}) \in \{\mathbf{0}\} \times \mathbb{F}_2^{k*} \\ C_v(\mathbf{a}) + 2W_u(\mathbf{a})(-1)^{\mathbf{1}\cdot\mathbf{a}} - 2^k, & \text{if } (\mathbf{a}, \mathbf{b}) \in U' \\ 2W_u(\mathbf{a})(-1)^{\mathbf{a}\cdot\mathbf{b}} + 2W_v(\mathbf{a} \oplus \mathbf{b})(-1)^{(\mathbf{a} \oplus \mathbf{1})\cdot\mathbf{b}} & \\ -2(-1)^{u(\mathbf{1} \oplus \mathbf{a} \oplus \mathbf{b})} + 2(-1)^{v(\mathbf{1} \oplus \mathbf{a}) \oplus u(\mathbf{1} \oplus \mathbf{a} \oplus \mathbf{b})} & \\ -2(-1)^{v(\mathbf{1} \oplus \mathbf{a})} + 2, & \text{if } (\mathbf{a}, \mathbf{b}) \in \mathbb{F}_2^{k*} \times \mathbb{F}_2^k \setminus U' \end{cases},$$

where $U' = \{(\mathbf{c}, \mathbf{c}) : \mathbf{c} \in \mathbb{F}_2^{k*}\} \subset \mathbb{F}_2^k \times \mathbb{F}_2^k$.

Proof. By the definition of the autocorrelation function, we immediately get that $C_f(\mathbf{0}, \mathbf{0}) = 2^n$. We now consider the values of $C_f(\mathbf{a}, \mathbf{b})$ for all $(\mathbf{a}, \mathbf{b}) \in \mathbb{F}_2^k \times \mathbb{F}_2^k \setminus \{(\mathbf{0}, \mathbf{0})\}$. Our discuss is mainly based on the fact that $\sum_{\mathbf{x} \in \mathbb{F}_2^k} (-1)^{\mathbf{c}\cdot\mathbf{x} \oplus \mathbf{d}\cdot\mathbf{y}}$ equals 0 if $\mathbf{c} \in \mathbb{F}_2^{k*}$ and equals 2^k otherwise, where $\mathbf{d}$ and $\mathbf{y}$ are arbitrary vectors in $\mathbb{F}_2^k$. We consider the values of $C_f(\mathbf{a}, \mathbf{b})$ for all $(\mathbf{a}, \mathbf{b}) \in \mathbb{F}_2^k \times \mathbb{F}_2^k \setminus \{(\mathbf{0}, \mathbf{0})\}$ from the following three cases:
Case (i): Let $(a, b) \in \{\mathbf{0}\} \times \mathbb{F}_2^{k*}$. In this case, we have

$$C_f(a, b) = \sum_{(\mathbf{x},\mathbf{y}) \in \{\mathbf{1}\} \times \mathbb{F}_2^k} (-1)^{f(\mathbf{1},\mathbf{y}) \oplus f(\mathbf{1},\mathbf{y} \oplus \mathbf{b})} + \sum_{(\mathbf{x},\mathbf{y}) \in T \times \mathbb{F}_2^k} (-1)^{f(\mathbf{x},\mathbf{y}) \oplus f(\mathbf{x},\mathbf{y} \oplus \mathbf{b})}$$

$$
\begin{aligned}
= & \sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{1},\mathbf{1}\oplus\mathbf{b}\}} (-1)^{u(\mathbf{y})\oplus u(\mathbf{y}\oplus b)} + (-1)^{v(\mathbf{1})\oplus u(\mathbf{1}\oplus b)} + (-1)^{u(\mathbf{1}\oplus\mathbf{b})\oplus v(\mathbf{1})} \\
& + \sum_{\mathbf{x}\in T}\Bigg(\sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{x},\mathbf{x}\oplus\mathbf{b}\}} (-1)^{(\mathbf{x}\oplus\mathbf{y})\cdot\mathbf{y}\oplus(\mathbf{x}\oplus\mathbf{y}\oplus\mathbf{b})\cdot(\mathbf{y}\oplus\mathbf{b})} + (-1)^{v(\mathbf{x})\oplus\mathbf{b}\cdot(\mathbf{x}\oplus\mathbf{b})} \\
& \quad +(-1)^{\mathbf{b}\cdot(\mathbf{x}\oplus\mathbf{b})\oplus v(\mathbf{x})}\Bigg)
\end{aligned}
$$

We now regroup terms by isolating the cross-correlation of u and its shift by $\mathbf{b}$. The sum over T is likewise reorganized into a constant component and a Walsh transform term of v. Simplifying each part using standard character sum identities, we finally express the sum as a combination of $C_u(\mathbf{b})$, $W_v(\mathbf{b})$, and signed terms involving $u(\mathbf{1})$, $v(\mathbf{1})$, and their shifted evaluations.

$$
\begin{aligned}
= & \sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{1},\mathbf{1}\oplus\mathbf{b}\}} (-1)^{u(\mathbf{y})\oplus u(\mathbf{y}\oplus b)} + 2(-1)^{v(\mathbf{1})\oplus u(\mathbf{1}\oplus b)} \\
& + \sum_{\mathbf{x}\in T}\Big((2^k-2)(-1)^{\mathbf{b}\cdot\mathbf{x}\oplus\mathbf{1}\cdot\mathbf{b}} + 2(-1)^{v(\mathbf{x})\oplus\mathbf{b}\cdot\mathbf{x}\oplus\mathbf{1}\cdot\mathbf{b}}\Big) \\
= & \; C_u(\mathbf{b}) - 2(-1)^{u(\mathbf{1})\oplus u(\mathbf{1}\oplus\mathbf{b})} + 2(-1)^{v(\mathbf{1})\oplus u(\mathbf{1}\oplus b)} \\
& +(2^k-2)\sum_{\mathbf{x}\in\mathbb{F}_2^k}(-1)^{\mathbf{b}\cdot\mathbf{x}\oplus\mathbf{1}\cdot\mathbf{b}} - (2^k-2) + 2W_v(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{b}} - 2(-1)^{v(\mathbf{1})} \\
= & \; C_u(\mathbf{b}) + 2W_v(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{b}} - 2(-1)^{u(\mathbf{1})\oplus u(\mathbf{1}\oplus\mathbf{b})} + 2(-1)^{v(\mathbf{1})\oplus u(\mathbf{1}\oplus b)} \\
& -2(-1)^{v(\mathbf{1})} - 2^k + 2,
\end{aligned}
$$

where $T = \mathbb{F}_2^k \setminus \{\mathbf{1}\}$. It can be easily verified that $v(\mathbf{1}) = 0$ if t is odd and $v(\mathbf{1}) = 1$ if t is even. In addition, we can check that $u(\mathbf{1}) = 0$. Thus, in this case, we have

$$
C_f(a,b) = C_u(\mathbf{b}) + 2W_v(\mathbf{b})(-1)^{\mathbf{1}\cdot\mathbf{b}} - 2^k + 2\big(1-(-1)^{t+1}\big)\big(1-(-1)^{u(\mathbf{1}\oplus\mathbf{b})}\big).
$$

Case (ii): Let $(\mathbf{a},\mathbf{b}) \in U'$. We can get that

$$
\begin{aligned}
C_f(a,b) = & \sum_{\mathbf{x}=\mathbf{1},\mathbf{y}\in\mathbb{F}_2^k} (-1)^{f(\mathbf{1},\mathbf{y})\oplus f(\mathbf{1}\oplus\mathbf{a},\mathbf{y}\oplus\mathbf{a})} + \sum_{\mathbf{x}=\mathbf{1}\oplus\mathbf{a},\mathbf{y}\in\mathbb{F}_2^k} (-1)^{f(\mathbf{1}\oplus\mathbf{a},\mathbf{y})\oplus f(\mathbf{1},\mathbf{y}\oplus\mathbf{a})} \\
& + \sum_{(\mathbf{x},\mathbf{y})\in E_\mathbf{a}\times\mathbb{F}_2^k} (-1)^{f(\mathbf{x},\mathbf{y})\oplus f(\mathbf{x}\oplus\mathbf{1},\mathbf{y}\oplus\mathbf{a})}.
\end{aligned}
$$

We simplify the expression by isolating special input cases and applying change-of-variable techniques to linearize the dot products. This reveals the Walsh transform of u and the cross-correlation of v, allowing the full sum to be expressed in a compact spectral form using character orthogonality and explicit correction terms.

$$
\begin{aligned}
= & \sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{1},\mathbf{1}\oplus\mathbf{a}\}} (-1)^{u(\mathbf{y})\oplus(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{y}\oplus\mathbf{a})\cdot(\mathbf{y}\oplus\mathbf{a})} + (-1)^{v(\mathbf{1})\oplus v(\mathbf{1}\oplus\mathbf{a})} + (-1)^{u(\mathbf{1}\oplus\mathbf{a})\oplus\mathbf{1}\cdot\mathbf{a}} \\
& + \sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{1},\mathbf{1}\oplus\mathbf{a}\}} (-1)^{(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{y})\cdot\mathbf{y}\oplus u(\mathbf{y}\oplus\mathbf{a})} + (-1)^{v(\mathbf{1}\oplus\mathbf{a})\oplus v(\mathbf{1})} + (-1)^{\mathbf{1}\cdot\mathbf{a}\oplus u(\mathbf{1}\oplus\mathbf{a})} \\
& + \sum_{\mathbf{x}\in E_a}\Bigg(\sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{x}\}} (-1)^{(\mathbf{x}\oplus\mathbf{y})\cdot\mathbf{y}\oplus(\mathbf{x}\oplus\mathbf{a}\oplus\mathbf{y}\oplus\mathbf{a})\cdot(\mathbf{y}\oplus\mathbf{a})} + (-1)^{v(\mathbf{x})\oplus v(\mathbf{x}\oplus\mathbf{a})}\Bigg) \\
= & \; 2\sum_{\mathbf{y}\in\mathbb{F}_2^k\setminus\{\mathbf{1},\mathbf{1}\oplus\mathbf{a}\}} (-1)^{u(\mathbf{y})\oplus\mathbf{a}\cdot\mathbf{y}\oplus\mathbf{1}\cdot\mathbf{a}} + (-1)^{v(\mathbf{1})\oplus v(\mathbf{1}\oplus\mathbf{a})} + (-1)^{u(\mathbf{1}\oplus\mathbf{a})\oplus\mathbf{1}\cdot\mathbf{a}}
\end{aligned}
$$

$$
\begin{aligned}
& + \sum_{\mathbf{x} \in E_a} \left(\sum_{\mathbf{y} \in \mathbb{F}_2^k} (-1)^{\mathbf{a}\cdot\mathbf{y} \oplus \mathbf{a}\cdot\mathbf{x}} - (-1)^{\mathbf{a}\cdot\mathbf{x} \oplus \mathbf{a}\cdot\mathbf{x}} + (-1)^{v(\mathbf{x}) \oplus v(\mathbf{x}\oplus\mathbf{a})} \right) \\
= & \left[2W_u(\mathbf{a})(-1)^{\mathbf{1}\cdot\mathbf{a}} - 2(-1)^{u(\mathbf{1})} + 2(-1)^{v(\mathbf{1})\oplus v(\mathbf{1}\oplus\mathbf{a})} \right] \\
& + \left[C_v(\mathbf{a}) - 2(-1)^{v(\mathbf{1})\oplus v(\mathbf{1}\oplus\mathbf{a})} - (2^k - 2) \right] \\
= & \ C_v(\mathbf{a}) + 2W_u(\mathbf{a})(-1)^{\mathbf{1}\cdot\mathbf{a}} - 2^k,
\end{aligned}
$$

where $E_{\mathbf{a}} = \mathbb{F}_2^k \setminus \{\mathbf{1}, \mathbf{1} \oplus \mathbf{a}\}$.
Case (iii): Let $(\mathbf{a}, \mathbf{b}) \in \mathbb{F}_2^{k*} \times \mathbb{F}_2^k \setminus U'$. We simplify $C_f(\mathbf{a}, \mathbf{b})$ by reorganizing terms to reveal Walsh transforms of u and v, using variable shifts and orthogonality.

$$
\begin{aligned}
C_f(\mathbf{a}, \mathbf{b}) = & \sum_{\mathbf{x}=\mathbf{1}, \mathbf{y} \in \mathbb{F}_2^k} (-1)^{f(\mathbf{1},\mathbf{y}) \oplus f(\mathbf{1}\oplus\mathbf{a}, \mathbf{y}\oplus\mathbf{b})} + \sum_{\mathbf{x}=\mathbf{1}\oplus\mathbf{a}, \mathbf{y} \in \mathbb{F}_2^k} (-1)^{f(\mathbf{1}\oplus\mathbf{a},\mathbf{y}) \oplus f(\mathbf{1}, \mathbf{y}\oplus\mathbf{b})} \\
& + \sum_{(\mathbf{x},\mathbf{y}) \in E_{\mathbf{a}} \times \mathbb{F}_2^k} (-1)^{f(\mathbf{x},\mathbf{y}) \oplus f(\mathbf{x}\oplus\mathbf{1}, \mathbf{y}\oplus\mathbf{b})} \\
= & \sum_{\mathbf{y} \in \mathbb{F}_2^k \setminus \{\mathbf{1}, \mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b}\}} \Big((-1)^{u(\mathbf{y}) \oplus (\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{y}\oplus\mathbf{b})\cdot(\mathbf{y}\oplus\mathbf{b})} + (-1)^{v(\mathbf{1}) \oplus (\mathbf{a}\oplus\mathbf{b})\cdot(\mathbf{1}\oplus\mathbf{b})} \\
& + (-1)^{u(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b}) \oplus v(\mathbf{1}\oplus\mathbf{a})} \Big) + \sum_{\mathbf{y} \in \mathbb{F}_2^k \setminus \{\mathbf{1}\oplus\mathbf{a}, \mathbf{1}\oplus\mathbf{b}\}} \Big((-1)^{(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{y})\cdot\mathbf{y} \oplus u(\mathbf{y}\oplus\mathbf{b})} \\
& + (-1)^{v(\mathbf{1}\oplus\mathbf{a}) \oplus u(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b})} + (-1)^{(\mathbf{a}\oplus\mathbf{b})\cdot(\mathbf{1}\oplus\mathbf{b}) \oplus v(\mathbf{1})} \Big) \\
& + \sum_{\mathbf{x} \in E_{\mathbf{a}}} \Big(\sum_{\mathbf{y} \in \mathbb{F}_2^k \setminus \{\mathbf{x}, \mathbf{x}\oplus\mathbf{a}\oplus\mathbf{b}\}} (-1)^{(\mathbf{x}\oplus\mathbf{y})\cdot\mathbf{y} \oplus (\mathbf{x}\oplus\mathbf{a}\oplus\mathbf{y}\oplus\mathbf{b})\cdot(\mathbf{y}\oplus\mathbf{b})} \\
& + (-1)^{v(\mathbf{x}) \oplus (\mathbf{a}\oplus\mathbf{b})\cdot(\mathbf{x}\oplus\mathbf{b})} + (-1)^{(\mathbf{a}\oplus\mathbf{b})\cdot(\mathbf{x}\oplus\mathbf{a}\oplus\mathbf{b}) \oplus v(\mathbf{x}\oplus\mathbf{a})} \Big) \\
= & \ 2 \sum_{\mathbf{y} \in \mathbb{F}_2^k \setminus \{\mathbf{1}, \mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b}\}} \Big((-1)^{u(\mathbf{y}) \oplus \mathbf{a}\cdot\mathbf{y} \oplus \mathbf{a}\cdot\mathbf{b}} + (-1)^{v(\mathbf{1}) \oplus \mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})} \\
& + (-1)^{v(\mathbf{1}\oplus\mathbf{a}) \oplus u(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b})} \Big) + \sum_{\mathbf{x} \in E_{\mathbf{a}}} \Big(\sum_{\mathbf{y} \in \mathbb{F}_2^k \setminus \{\mathbf{x}, \mathbf{x}\oplus\mathbf{a}\oplus\mathbf{b}\}} (-1)^{\mathbf{a}\cdot\mathbf{y} \oplus (\mathbf{a}\oplus\mathbf{1}\oplus\mathbf{x})\cdot\mathbf{b}} \\
& + (-1)^{v(\mathbf{x}) \oplus (\mathbf{a}\oplus\mathbf{b})\cdot\mathbf{x} \oplus \mathbf{a}\cdot\mathbf{b} \oplus \mathbf{1}\cdot\mathbf{b}} + (-1)^{v(\mathbf{x}\oplus\mathbf{a}) \oplus (\mathbf{a}\oplus\mathbf{b})\cdot\mathbf{x} \oplus \mathbf{1}\cdot(\mathbf{a}\oplus\mathbf{b})} \Big).
\end{aligned}
$$

In this step, we partition the summation domain by excluding special points such as $\mathbf{1}$, $\mathbf{1} \oplus \boldsymbol{\alpha}$, and $\mathbf{1} \oplus \boldsymbol{\alpha} \oplus \boldsymbol{\beta}$. Applying change of variables and exploiting the orthogonality of characters, the nested exponent terms involving dot products are transformed into expressions involving the Walsh transforms $W_u(\boldsymbol{\alpha})$ and $W_v(\boldsymbol{\alpha} \oplus \boldsymbol{\beta})$. The remaining exceptional cases are handled explicitly, resulting in a decomposition of $C_f(\boldsymbol{\alpha}, \boldsymbol{\beta})$ into spectral components plus correction terms.

$$
\begin{aligned}
= & \ 2 \sum_{\mathbf{y} \in \mathbb{F}_2^k \setminus \{\mathbf{1}, \mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b}\}} \Big((-1)^{u(\mathbf{y}) \oplus \mathbf{a}\cdot\mathbf{y} \oplus \mathbf{a}\cdot\mathbf{b}} + (-1)^{v(\mathbf{1}) \oplus \mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})} \\
& + (-1)^{v(\mathbf{1}\oplus\mathbf{a}) \oplus u(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b})} \Big) + \sum_{\mathbf{x} \in E_{\mathbf{a}}} \Big(- (-1)^{(\mathbf{a}\oplus\mathbf{b})\cdot\mathbf{x} \oplus \mathbf{b}\cdot(\mathbf{1}\oplus\mathbf{a})}
\end{aligned}
$$

$$
\begin{aligned}
& -(-1)^{(\mathbf{a}\oplus\mathbf{b})\cdot\mathbf{x}\oplus\mathbf{1}\cdot(\mathbf{a}\oplus\mathbf{b})}+(-1)^{v(\mathbf{x})\oplus(\mathbf{a}\oplus\mathbf{b})\cdot\mathbf{x}\oplus\mathbf{a}\cdot\mathbf{b}\oplus\mathbf{1}\cdot\mathbf{b}} \\
& +(-1)^{v(\mathbf{x}\oplus\mathbf{a})\oplus(\mathbf{a}\oplus\mathbf{b})\cdot\mathbf{x}\oplus\mathbf{1}\cdot(\mathbf{a}\oplus\mathbf{b})}\Big) \\
= \; & \big(2W_u(\mathbf{a})(-1)^{\mathbf{a}\cdot\mathbf{b}}-2(-1)^{u(\mathbf{1})\oplus\mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})}-2(-1)^{u(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b})} \\
& +2(-1)^{v(\mathbf{1})\oplus\mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})}+2(-1)^{v(\mathbf{1}\oplus\mathbf{a})\oplus u(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b})}\big) \\
& +\big(2W_v(\mathbf{a}\oplus\mathbf{b})(-1)^{(\mathbf{a}\oplus\mathbf{1})\cdot\mathbf{b}}-2(-1)^{v(\mathbf{1})\oplus\mathbf{a}\cdot(\mathbf{b}\oplus\mathbf{1}_k)} \\
& -2(-1)^{v(\mathbf{1}\oplus\mathbf{a})}+2(-1)^{\mathbf{a}\cdot(\mathbf{1}\oplus\mathbf{b})}+2\big) \\
= \; & 2W_u(\mathbf{a})(-1)^{\mathbf{a}\cdot\mathbf{b}}+2W_v(\mathbf{a}\oplus\mathbf{b})(-1)^{(\mathbf{a}\oplus\mathbf{1})\cdot\mathbf{b}}-2(-1)^{u(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b})} \\
& +2(-1)^{v(\mathbf{1}\oplus\mathbf{a})\oplus u(\mathbf{1}\oplus\mathbf{a}\oplus\mathbf{b})}-2(-1)^{v(\mathbf{1}\oplus\mathbf{a})}+2,
\end{aligned}
$$

where $E_{\mathbf{a}}=\mathbb{F}_2^k\setminus\{\mathbf{1},\mathbf{1}\oplus\mathbf{a}\}$. □

From the above results, we can derive the maximum absolute autocorrelation value of the constructed balanced function in $2k$ variables, which is strictly less than 2^k.

Theorem 3. *Let $n=2k=4t\geq 20$ and $f\in\mathcal{B}_n$ be a Boolean function generated by Construction 1. Then the absolute indicator of $f\in\mathcal{B}_n$ is*

$$\Delta_f\leq 2^k-2^{k-2}+9\cdot 2^t.$$

Proof. If $\mathbf{a}=\mathbf{b}=\mathbf{0}\in\mathbb{F}_2^k$, then $C_f(\mathbf{a},\mathbf{b})=2^n$. We derive the maximum absolute autocorrelation value of f on different cases.
Case(i): Let us consider $\mathbf{a}=\mathbf{b}\neq\mathbf{0}\in\mathbb{F}_2^k$. Suppose $\mathbf{a}=(\mathbf{c},\mathbf{d})\in\mathbb{F}_2^t\times\mathbb{F}_2^t\setminus\{(\mathbf{0},\mathbf{0})\}$. Then

$$
\begin{aligned}
C_f(\mathbf{a},\mathbf{a}) &= C_v(\mathbf{c},\mathbf{d})+2W_u(\mathbf{c},\mathbf{d})(-1)^{wt(\mathbf{c},\mathbf{d})}-2^k \\
&= \begin{cases} C_v(\mathbf{c},\mathbf{0})+2W_u(\mathbf{c},\mathbf{0})(-1)^{wt(\mathbf{c})}-2^k, & \text{if } \mathbf{c}\neq\mathbf{0},\mathbf{d}=\mathbf{0} \\ C_v(\mathbf{c},\mathbf{d})+2W_u(\mathbf{c},\mathbf{d})(-1)^{wt(\mathbf{c},\mathbf{d})}-2^k, & \text{if } \mathbf{d}\neq\mathbf{0} \end{cases}.
\end{aligned}
$$

Since $-2^t\leq 2W_u(\mathbf{c},\mathbf{0})(-1)^{wt(\mathbf{c})}\leq 2^t$ for $\mathbf{c}\neq\mathbf{0}$ and $-3\cdot 2^t\leq 2W_u(\mathbf{c},\mathbf{d})(-1)^{wt(\mathbf{c},\mathbf{d})}\leq 3\cdot 2^t$ for $\mathbf{d}\neq\mathbf{0}$. From Lemma 8, we have $\max_{\mathbf{a}\in\mathbb{F}_2^k\setminus\{\mathbf{0}\}}|C_f(\mathbf{a},\mathbf{a})|\leq 2^k-2^{k-2}+9\cdot 2^t$.
Case(ii): Let $\mathbf{a}=\mathbf{0}$ and $\mathbf{b}\neq\mathbf{0}\in\mathbb{F}_2^k$. Suppose $\mathbf{b}=(\mathbf{e},\mathbf{h})\in\mathbb{F}_2^t\times\mathbb{F}_2^t\setminus\{(\mathbf{0},\mathbf{0})\}$. Then

$$
\begin{aligned}
C_f(\mathbf{0},\mathbf{b}) &= C_u(\mathbf{b})+2W_v(\mathbf{b})(-1)^{wt(\mathbf{b})}-2^k+2 \\
&\quad -2(-1)^{u(\mathbf{1})\oplus u(\mathbf{1}\oplus\mathbf{b})}+2(-1)^{v(\mathbf{1})\oplus u(\mathbf{1}\oplus\mathbf{b})}-2(-1)^{v(\mathbf{1})} \\
&= A(\mathbf{b})+B(\mathbf{b}),
\end{aligned}
$$

where $A(\mathbf{b})=C_u(\mathbf{b})+2W_v(\mathbf{b})(-1)^{wt(\mathbf{b})}-2^k$ and $B(\mathbf{b})=2-2(-1)^{u(\mathbf{1})\oplus u(\mathbf{1}\oplus\mathbf{b})}$ $+2(-1)^{v(\mathbf{1})\oplus u(\mathbf{1}\oplus\mathbf{b})}-2(-1)^{v(\mathbf{1})}$ for all $\mathbf{b}\in\mathbb{F}_2^k$. Since $\max_{\mathbf{b}\in\mathbb{F}_2^{k*}}|B(\mathbf{b})|\leq 8$ and

$$
\begin{aligned}
A(\mathbf{b}) &= C_u(\mathbf{e},\mathbf{h})+2W_v(\mathbf{e},\mathbf{h})(-1)^{wt(\mathbf{e},\mathbf{h})}-2^k \\
&= \begin{cases} C_u(\mathbf{e},\mathbf{0})+2W_v(\mathbf{e},\mathbf{0})(-1)^{wt(\mathbf{e})}-2^k, & \text{if } \mathbf{e}\neq\mathbf{0},\mathbf{h}=\mathbf{0} \\ C_u(\mathbf{e},\mathbf{h})+2W_v(\mathbf{e},\mathbf{h})(-1)^{wt(\mathbf{e},\mathbf{h})}-2^k, & \text{if } \mathbf{h}\neq\mathbf{0} \end{cases} \\
&= \begin{cases} C_u(\mathbf{e},\mathbf{0})-2^t-2^k, & \text{if } \mathbf{e}\neq\mathbf{0}, wt(\mathbf{e}) \text{ is even}, \mathbf{h}=\mathbf{0} \\ C_u(\mathbf{e},\mathbf{0})+2^t+2^k, & \text{if } \mathbf{e}\neq\mathbf{0}, wt(\mathbf{e}) \text{ is odd}, \mathbf{h}=\mathbf{0} \\ C_u(\mathbf{e},\mathbf{h})+2W_v(\mathbf{e},\mathbf{h})(-1)^{wt(\mathbf{e},\mathbf{h})}-2^k, & \text{if } \mathbf{h}\neq\mathbf{0} \end{cases}.
\end{aligned}
$$

Thus from Lemma 8, we have $\max|A(\mathbf{b})|\leq 2^k-2^{k-2}+2^{t+1}$ for $\mathbf{b}=(\mathbf{e},\mathbf{0})$ with $\mathbf{e}\neq\mathbf{0}$. If $\mathbf{h}\neq\mathbf{0}$, then we have

$$\max|A(\mathbf{b})|\leq\begin{cases} 2^k-2^{k-2}+4\cdot 2^t+2^{\frac{t+3}{2}}, & \text{if } t \text{ is odd} \\ 2^k-2^{k-2}+4\cdot 2^t+2^{\frac{t}{2}+2}, & \text{if } t \text{ is even} \end{cases},$$

and so, $\max_{\mathbf{b}\in\mathbb{F}_2^{k*}} |C_f(\mathbf{0},\mathbf{b})| \leq \begin{cases} 2^k - 2^{k-2} + 2^{t+2} + 2^{\frac{t+3}{2}} + 8, & \text{if } t \text{ is odd} \\ 2^k - 2^{k-2} + 2^{t+2} + 2^{\frac{t}{2}+2} + 8, & \text{if } t \text{ is even} \end{cases}$.

Case(iii): Let $\mathbf{a} \neq \mathbf{0}$, $\mathbf{b} \neq \mathbf{0}$ with $\mathbf{a} \neq \mathbf{b}$. Then we have

$$\max_{\mathbf{a},\mathbf{b}\in\mathbb{F}_2^{k*},\mathbf{a}\neq\mathbf{b}} |C_f(\mathbf{a},\mathbf{b})| \leq \begin{cases} 3\cdot 2^{t+1} + 2^{k-2} + 2^{\frac{t+3}{2}} + 8, & \text{if } t \text{ is odd} \\ 3\cdot 2^{t+1} + 2^{k-2} + 2^{\frac{t}{2}+2} + 8, & \text{if } t \text{ is even} \end{cases}.$$

Combining all three cases, we get the result. □

3.3.4 Algebraic Degree

Finally, we show that the algebraic degree of the balanced Boolean function defined in Construction 1 is high. We formally state and prove the following bound on the algebraic degree of the constructed function.

Theorem 4. *Let $n = 2k = 4t$ be an even integer not less than 20 and t be an even integer. The algebraic degree of n-variable Boolean function $f \in \mathcal{B}_n$ generated by Construction 1 is*

$$\deg(f) = \frac{3n}{4} + 2.$$

Proof. We recall that the outputs of the MM function $(\mathbf{x} \oplus \mathbf{y}) \cdot \mathbf{y} = \mathbf{x} \cdot \mathbf{y} \oplus \bigoplus_{i=1}^{k} y_i$ are modified by a sub-function u when $\mathbf{x} = \mathbf{1}$ and $\mathbf{y} \neq \mathbf{1}$, and by a sub-function v when $\mathbf{x} = \mathbf{y}$, where $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^k$. The function f can be written as

$$\begin{aligned} f(\mathbf{x},\mathbf{y}) = {} & (x_1x_2\cdots x_k(y_1y_2\cdots y_k \oplus 1) \oplus 1)(\mathbf{x}\oplus\mathbf{y})\cdot\mathbf{y} \\ & \oplus x_1x_2\cdots x_k(y_1y_2\cdots y_k \oplus 1)u(\mathbf{y}) \\ & \oplus (x_1 \oplus y_1 \oplus 1)(x_2 \oplus y_2 \oplus 1)\cdots(x_k \oplus y_k \oplus 1)v(\mathbf{x}). \end{aligned}$$

The algebraic degree of the balanced Boolean functions defined in Construction 1 depends on the sub-functions u and v. According to Definition 3, we have $\deg(u) = 4$ and $\deg(v) = t+2$. Substituting these values into the final expression for the function yields the desired bound on the algebraic degree of the constructed function, namely, $3t + 2 = \frac{3n}{4} + 2$.

□

3.4 Circuit Realization of Construction 1

Recall that the constructed Boolean function is defined as an n-variable function f over $\mathbb{F}_2^k \times \mathbb{F}_2^k$ as follows:

$$f(\mathbf{x},\mathbf{y}) = \begin{cases} u(\mathbf{y}), & \text{if } \mathbf{x} = \mathbf{1},\ \mathbf{y} \in \mathbb{F}_2^k \setminus \{\mathbf{1}\}, \\ v(\mathbf{x}), & \text{if } \mathbf{x} = \mathbf{y} \in \mathbb{F}_2^k, \\ (\mathbf{x}\oplus\mathbf{y})\cdot\mathbf{y}, & \text{otherwise}, \end{cases}$$

where u and v are Boolean functions over $\mathbb{F}_2^k$, as defined in Definition 3.

We now describe how Fig. 3 implements the function f using logic circuits. The circuit consists of two 2×1 multiplexers and uses two indicator functions, h_1 and h_2, which serve as the selector lines for these multiplexers. These functions are defined as follows:

- $h_1(\mathbf{x},\mathbf{y})$ outputs 0 if $\mathbf{x} = \mathbf{1}$ and $\mathbf{y} \neq \mathbf{1}$,
- $h_2(\mathbf{x},\mathbf{y})$ outputs 0 if $\mathbf{x} = \mathbf{y}$, for $\mathbf{x},\mathbf{y} \in \mathbb{F}_2^k$.

Let $\varepsilon \in \mathbb{F}_2$ denote the output of the original MM bent function, that is, $(\mathbf{x} \oplus \mathbf{y}) \cdot \mathbf{y}$. The output after the first multiplexer is given by:

$$\varepsilon_1 = (1 \oplus h_1(\mathbf{x}, \mathbf{y}))u(\mathbf{y}) \oplus h_1(\mathbf{x}, \mathbf{y})\varepsilon.$$

Finally, the overall output of the circuit after the second multiplexer is:

$$f(\mathbf{x}, \mathbf{y}) = (1 \oplus h_2(\mathbf{x}, \mathbf{y}))v(\mathbf{x}) \oplus h_2(\mathbf{x}, \mathbf{y})\varepsilon_1.$$

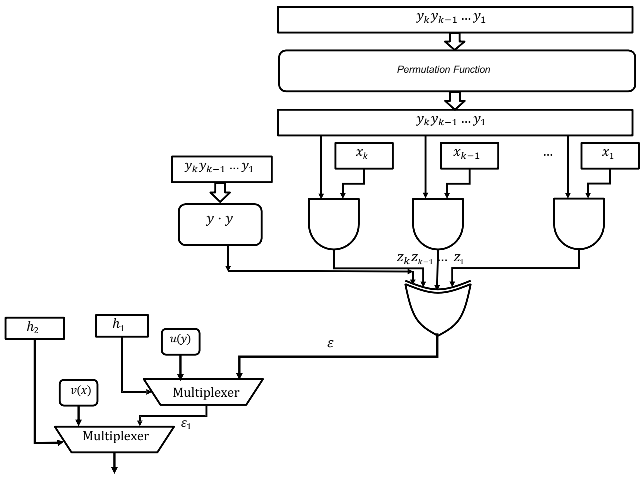

Figure 3: Balanced function given in Construction 1

We now analyze the circuit realization cost of the constructed Boolean function. The indicator functions h_1 and h_2 can be implemented as:

$$h_1(\mathbf{x}, \mathbf{y}) = x_1 x_2 \cdots x_k \oplus 1,$$

$$h_2(\mathbf{x}, \mathbf{y}) = (x_1 \oplus y_1) \vee (x_2 \oplus y_2) \vee \cdots \vee (x_k \oplus y_k).$$

It is straightforward to verify that the implementations of h_1 and h_2 require k and $2k$ logic gates, respectively. The computation of $\mathbf{y} \cdot \mathbf{y} = y_1 \oplus y_2 \oplus \cdots \oplus y_k$ involves only k XOR gates. According to Definition 3, the sub-functions u and v require $2k$ and $3k + \frac{k}{2} + 1$ logic gates, respectively. For $k = 2t \geq 10$, the total number of logic gates required to implement all the constituent components sums to approximately $9.5k$.

In addition, computing the scalar product $\mathbf{x} \cdot \mathbf{y}$ incurs a cost of $2k - 1$ logic gates. The two multiplexers used in the circuit contribute an additional 8 logic gates (comprising 4 AND gates and 4 XOR gates). Consequently, the total number of logic gates required to implement the balanced Boolean function f in $2k \geq 20$ variables is:

$$11.5k + 8,$$

which is upper bounded by $12k$, or equivalently $6n$, where $n = 2k$ denotes the total number of input variables.

Remark 2. Furthermore, the Boolean function defined in Construction 1 and illustrated in Fig. 3 exhibits low multiplicative complexity. Specifically, AND gates are required to implement the components $\mathbf{x} \cdot \mathbf{y}$, h_1, u, v, and the multiplexers, where $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^k$. The AND gate requirements for each component are as follows:

- k AND gates for computing the scalar product $\mathbf{x} \cdot \mathbf{y}$,
- $k - 1$ AND gates for realizing the function h_1,
- $k + 1$ AND gates for implementing the function u,
- $3t + \iota + 2$ AND gates for realizing the function v, where $k = 2t$ and $\iota = \left\lfloor \frac{t-1}{2} \right\rfloor$,
- 4 AND gates for implementing the multiplexers.

Combining these, the total number of AND gates required to implement the Boolean function $f \in \mathcal{B}_{2k}$ as depicted in Fig. 3 is given by:

$$3k + t + \iota + 6,$$

where $k = 2t$ and $\iota = \left\lfloor \frac{t-1}{2} \right\rfloor$. This demonstrates that the function f achieves linear multiplicative complexity with respect to the number of input variables $n = 2k$.

4 Applications of our construction

Our primary objective is to demonstrate that specific subclasses of MM-type Boolean functions can be realized using small size circuits composed of only standard logic gates of fan-in 2. Notably, the core construction we propose requires only a linear number of logic gates, specifically, fewer than $6n$ gates for n input variables. This class of functions exhibits strong cryptographic characteristics, particularly with respect to nonlinearity and autocorrelation. Thus such functions can be used as nonlinear combiner in different stream cipher models.

Although various generalizations of MM-type constructions have been introduced in the literature, disciplined approaches for implementing such functions with small size circuits remain scarce. In fact, recent works such as [KCMM18, TKMM19] employ exponentially many gates in terms of the number of input variables to realize Boolean functions with desirable cryptographic properties. From the perspective of circuit realization, we introduce a novel construction that achieves exceptional cryptographic parameters while maintaining implementation efficiency, an approach not previously attempted. Specifically, we identify a subclass of Boolean functions that not only retain strong cryptographic characteristics but are also amenable to small size circuit realization. We then provide a theoretical framework for constructing such functions over a large number of variables in a lightweight and structured manner.

Potential applications of this function class may include Fully Homomorphic Encryption (FHE)-friendly stream cipher designs and enhanced resistance against Differential Fault Attacks (DFA). To demonstrate the practical impact, we explain how one can incorporate our proposed functions into well-known stream ciphers by replacing their core Boolean functions with instances derived from our construction. It is needless to mention that the functions used are either on larger number of variables, or significantly more complex, thereby having much better properties related to confusion and diffusion. Further, preliminary observations suggest that the the existing DFAs cannot work immediately once our newly designed functions are incorporated. Nonetheless, further cryptanalytic evaluation is warranted to fully assess their resilience and performance.

4.1 Modification of FLIP

Symmetric-key cryptographic primitives are used in practical hybrid Fully Homomorphic Encryption (FHE) schemes [Gen09, RAD78]. In such schemes, the main computational bottleneck arises from the multiplicative complexity of symmetric primitives, rendering traditional standards like AES and SHA-3 inefficient. This has led to a growing interest in FHE-friendly symmetric-key primitives. It is important to note that such functions are typically defined over a large number of variables, often around 500. Many existing constructions of cryptographic Boolean functions are inadequate in this regime due to high implementation cost or the lack of explicit constructions. Hence, the design of new primitives and paradigms that simultaneously satisfy cryptographic strength and low circuit realization cost has become an active area of research—one which can benefit from the efficient implementation technique described here.

As an illustration, consider the filter function of FLIP $(42, 128,^8 \Delta^9)$, which is the direct sum of three distinct Boolean functions:

$$f = f_1 \oplus f_2 \oplus f_3, \text{ where}$$

- f_1 is a linear function of the first 42 variables,
- f_2 is a quadratic Boolean function in the next 128 variables,
- f_3 is the sum of triangular functions with 45 variables each.

Suppose $\mathbf{x}, \mathbf{y} \in \mathbb{F}_2^{180}$ and $\mathbf{x} = (\mathbf{x}', \mathbf{x}'')$, $\mathbf{y} = (\mathbf{y}', \mathbf{y}'')$ with $\mathbf{x}', \mathbf{x}'', \mathbf{y}', \mathbf{y}'' \in \mathbb{F}_2^{90}$. Let the state vector of FLIP $(42, 128,^8 \Delta^9)$ be $(s_0, s_1, \ldots, s_{529})$, and define:

$$x_i = s_{169+i}, \quad y_i = s_{349+i}, \quad i = 1, 2, \ldots, 180.$$

Following Construction 1, we can update the triangular function using our candidate function:

$$T_{\text{updat}}(\mathbf{x}, \mathbf{y}) = \begin{cases} u(\mathbf{y}), & \text{if } \mathbf{x} = \mathbf{1},\ \mathbf{y} \in \mathbb{F}_2^{180} \setminus \{\mathbf{1}\}, \\ v(\mathbf{x}), & \text{if } \mathbf{y} = \mathbf{x} \in \mathbb{F}_2^{180}, \\ (\mathbf{x} \oplus \mathbf{y}) \cdot \mathbf{y}, & \text{otherwise}, \end{cases}$$

where $u, v \in \mathcal{B}_{180}$ are defined as:

$$\begin{aligned} u(\mathbf{y}', \mathbf{y}'') &= \left(\bigoplus_{i=1}^{90} y_i y_{90+i} \right) \left(\bigoplus_{i=2}^{90} y_i y_{89+i} \oplus y_1 y_{180} \oplus y_2 y_{180} \right), \\ v(\mathbf{x}', \mathbf{x}'') &= \left(\bigoplus_{i=1}^{44} x_{40+i} x_{59+i} \oplus x_{179} \oplus x_{180} \right) \prod_{i=1}^{90} (x_i \oplus 1) \\ &\quad \oplus \left(\bigoplus_{i=1}^{90} x_i x_{90+i} \oplus 1 \right) \left(\bigoplus_{i=2}^{90} x_i x_{79+i} \oplus x_1 x_{80} \oplus x_2 x_{80} \right). \end{aligned}$$

The nonlinearity of T_{updat} is

$$2^{359} - 2^{179} - 3 \times 2^{89} - 2^{45},$$

which is significantly higher than that of triangular functions. Furthermore, the absolute indicator satisfies:

$$\Delta_{T_{\text{updat}}} \leq 2^{180} - 2^{178} + 9 \times 2^{90}.$$

The function T_{updat} can be implemented as shown in Figure 3. For the circuit realization of T_{updat}, the total number of AND gates required is:

$$3 \times 180 + 90 + 44 + 6 = 680.$$

Using these balanced Boolean functions in the filter function of FLIP $(42, 128,^8 \Delta^9)$, we can achieve improved cryptographic properties. With this modification, the filter function f gains significantly better security properties, albeit at the cost of a larger circuit.

Moreover, cryptanalytic attacks against such a modified model must be revisited. In particular, we observed that the existing Differential Fault Attack against FLIP [RBM21, Section 4] (by utilizing the SAT solver in SageMath on a similar hardware) fails to produce a result even after two hours with a single fault when the filter function is replaced by our improved construction.

4.2 Modification of Grain v1

Such functions have immediate applications in resisting Differential Fault Attacks (DFAs) in stream ciphers based on Feedback Shift Registers (FSRs). The most critical step in DFAs involves exploiting both normal and faulty outputs to retrieve key information. Typically, attackers construct algebraic equations involving the key and attempt to solve them [SBM15, RBM21]. Common solving techniques include linearization, Gröbner basis computation, and equation solvers like SAT solvers. However, equations with higher algebraic degree and increased complexity are generally more difficult to solve. Hence, incorporating Boolean functions with higher algebraic degree can enhance a cipher's resistance to DFAs. Before recovering key information, attackers must also determine the fault location. Employing functions with good autocorrelation properties helps prevent signature-based fault identification, thereby strengthening DFA resistance. For instance, consider stream ciphers in the Grain family. Although the total FSR lengths are substantial (160 bits in Grain v1 and 256 bits in Grain-128a), only a few bits (5 and 9, respectively) are used in the combining function. This limited usage facilitates signature extraction and equation solving, as demonstrated in [SBM15].

On the other hand, simply increasing the size of the Boolean function is not necessarily effective. For example, in the FLIP cipher, the filter function involves 200 to 400 variables and spans all FSR bits. Despite this, due to inadequate cryptographic properties, the resulting equations can still be efficiently solved [RBM21]. We propose using the balanced Boolean functions from Construction 1 to design a stream cipher. As a case study, we consider a modification of the filter function in the Grain v1 stream cipher (which uses 80-bit keys). The core components of Grain v1 are an LFSR and an NFSR, each of length 80, and an output function over 160 variables. The states of the LFSR and NFSR at time t are denoted by:

$$S_t = s_t, s_{t+1}, \dots, s_{t+79}, \quad B_t = b_t, b_{t+1}, \dots, b_{t+79}.$$

The cipher is initialized with an 80-bit key K and a 64-bit initial value (IV). Denoting the key bits by k_i ($0 \le i \le 79$) and IV bits by IV_i ($0 \le i \le 63$), the initial state is loaded as follows:

$$\begin{cases} b_i = k_i, & 0 \le i \le 79, \\ s_i = IV_i, & 0 \le i \le 63, \\ s_i = 1, & 64 \le i \le 79. \end{cases}$$

The update functions for the LFSR and NFSR remain the same as in the original Grain cipher:

$$\begin{aligned} s_{t+80} &= s_{t+62} \oplus s_{t+51} \oplus s_{t+38} \oplus s_{t+23} \oplus s_{t+13} \oplus s_t, \\ b_{t+80} &= s_t \oplus b_{t+62} \oplus b_{t+60} \oplus b_{t+52} \oplus b_{t+45} \oplus b_{t+37} \oplus b_{t+33} \oplus b_{t+28} \\ &\quad \oplus b_{t+21} \oplus b_{t+14} \oplus b_{t+9} \oplus b_t \oplus b_{t+63}b_{t+60} \oplus b_{t+37}b_{t+33} \oplus b_{t+15}b_{t+9} \\ &\quad \oplus b_{t+60}b_{t+52}b_{t+45} \oplus b_{t+33}b_{t+28}b_{t+21} \oplus b_{t+63}b_{t+45}b_{t+28}b_{t+9} \\ &\quad \oplus b_{t+60}b_{t+52}b_{t+37}b_{t+33} \oplus b_{t+63}b_{t+60}b_{t+21}b_{t+15} \\ &\quad \oplus b_{t+63}b_{t+60}b_{t+52}b_{t+45}b_{t+37} \oplus b_{t+33}b_{t+28}b_{t+21}b_{t+15}b_{t+9} \\ &\quad \oplus b_{t+52}b_{t+45}b_{t+37}b_{t+33}b_{t+28}b_{t+21}. \end{aligned}$$

The output function uses all 160 bits from the LFSR and NFSR. Let $x_i = s_{i-1}$ and $y_i = b_{i-1}$ for $1 \le i \le 80$. Then define

$$\mathbf{x} = (\mathbf{x}', \mathbf{x}'') = (x_1, \ldots, x_{40}, x_{41}, \ldots, x_{80}), \quad \mathbf{y} = (\mathbf{y}', \mathbf{y}'') = (y_1, \ldots, y_{40}, y_{41}, \ldots, y_{80}) \in \mathbb{F}_2^{80}.$$

The output function $h(\mathbf{x}, \mathbf{y})$ is defined as:

$$h(\mathbf{x}, \mathbf{y}) = \begin{cases} u(\mathbf{y}), & \text{if } \mathbf{x} = \mathbf{1},\ \mathbf{y} \neq \mathbf{1}, \\ v(\mathbf{x}), & \text{if } \mathbf{x} = \mathbf{y}, \\ (\mathbf{x} \oplus \mathbf{y}) \cdot \mathbf{y}, & \text{otherwise,} \end{cases}$$

where $u, v \in \mathcal{B}_{80}$ are given by:

$$\begin{aligned} u(\mathbf{y}', \mathbf{y}'') &= \left(\bigoplus_{i=1}^{40} y_i y_{40+i} \right) \left(\bigoplus_{i=2}^{40} y_i y_{39+i} \oplus y_1 y_{80} \oplus y_2 y_{80} \right), \\ v(\mathbf{x}', \mathbf{x}'') &= \left(\bigoplus_{i=1}^{19} x_{40+i} x_{59+i} \oplus x_{79} \oplus x_{80} \right) \prod_{i=1}^{40} (x_i \oplus 1) \\ &\quad \oplus \left(\bigoplus_{i=1}^{40} x_i x_{40+i} \oplus 1 \right) \left(\bigoplus_{i=2}^{40} x_i x_{39+i} \oplus x_1 x_{80} \oplus x_2 x_{80} \right). \end{aligned}$$

The nonlinearity of h is

$$2^{159} - 2^{79} - 3 \cdot 2^{39} - 2^{20},$$

and its absolute indicator satisfies

$$\Delta_h \le 2^{80} - 2^{78} + 9 \cdot 2^{40}.$$

We implement the function h as shown in Figure 3. The total number of AND gates required to realize h is

$$3 \cdot 80 + 40 + 19 + 6 = 305.$$

In [SBM15], a DFA using 10 fault injections was proposed. With our construction, signature extraction (as in [SBM15, Page 1652]) fails due to the extremely low autocorrelation of the function h. Additionally, considering that the fault locations are known, the attempt to solve the resulting equations using the same fault model from [SBM15] with SAT solvers did not produce any result, owing to the much higher complexity of the combining function. While the original Grain v1 used very few logic gates in its output function, our design, although requiring more logic gates, is still within practical bounds and significantly strengthens resistance against DFAs.

5 Conclusion

In this work, we present a generic construction technique to obtain cryptographically significant balanced Boolean functions with linear-size circuit requirement, achieved through a systematic modification of the classical Maiorana-McFarland (MM) framework. The resulting functions exhibit high nonlinearity and very low absolute autocorrelation, making them well-suited for cryptographic applications. This class of functions offers a rich design space with tunable trade-offs between cryptographic strength and implementation efficiency, thereby expanding the toolkit available to cryptographic system designers. Looking ahead, natural extensions of this work include achieving additional cryptographic criteria such as correlation immunity without significantly increasing the circuit complexity, minimizing the overall gate count further, and designing concrete FHE and DFA-resistant cipher instances based on the proposed functions. It would also be worthwhile to study the realizability of such functions under alternative gate models and logic synthesis constraints.

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References

[Car10] C. Carlet. Boolean functions for cryptography and error correcting codes. In Y. Crama and P. Hammer, editors, *Boolean Models and Methods*, pages 257–397. Cambridge Univ. Press, Cambridge, 2010. doi:10.1017/9781108606806.006.

[CMY+18] A. Canteaut, S. Maitra, H. Yoshida, L. Perrin, S. Jha, R. Rohit, and A. Baksi. Design of filtering functions for lightweight stream ciphers, 2018. Presentation at ASK 2018, Kolkata, India.

[CS09] T. W. Cusick and P. Stănică. *Cryptographic Boolean Functions and Applications*. Elsevier–Academic Press, 2009. doi:10.1016/C2016-0-00852-5.

[Dil74] J. F. Dillon. *Elementary Hadamard Difference Sets*. PhD thesis, University of Maryland, 1974. doi:10.13016/M2MS3K194.

[Dob94] H. Dobbertin. Construction of bent functions and balanced boolean functions with high nonlinearity. In *Fast Software Encryption*, volume 1008 of *LNCS*, pages 61–74, 1994. doi:10.1007/3-540-60590-8_5.

[Gen09] Craig Gentry. Fully homomorphic encryption using ideal lattices. In *Proceedings of the 41st Annual ACM Symposium on Theory of Computing*, pages 169–178, 2009. doi:10.1145/1536414.1536440.

[GS04] K. C. Gupta and P. Sarkar. Efficient representation and software implementation of resilient maiorana–mcfarland s-boxes. In *WISA*, volume 3325 of *LNCS*, pages 317–331, 2004. doi:10.1007/978-3-540-31815-6_26.

[KCMM18] M. Khairallah, A. Chattopadhyay, B. Mandal, and S. Maitra. On hardware implementation of tang–maitra boolean functions. In *Arithmetic of Finite Fields – 7th International Workshop, WAIFI*, volume 11321 of *LNCS*, pages 111–127, 2018. doi:10.1007/978-3-030-05153-2_6.

[KMT19] S. Kavut, S. Maitra, and D. Tang. Construction and search of balanced boolean functions on even number of variables towards excellent autocorrelation profile. *Designs, Codes and Cryptography*, 87(2–3):261–276, 2019. doi:10.1007/s10623-018-0522-1.

[Mes16] S. Mesnager. *Bent Functions – Fundamentals and Results*. Springer, Switzerland, 2016. doi:10.1007/978-3-319-32595-8.

[OKK23] E. Ozcekic, S. Kavut, and H. Kutucu. Genetic approach to improve cryptographic properties of balanced boolean functions using bent functions. *Computers*, 12(8):159, 2023. doi:10.3390/computers12080159.

[RAD78] R. L. Rivest, L. Adleman, and M. L. Dertouzos. On data banks and privacy homomorphisms. *Foundations of Secure Computation*, 4(11):169–180, 1978. URL: https://luca-giuzzi.unibs.it/corsi/Support/papers-cryptography/RAD78.pdf.

[RBM21] D. Roy, B. N. Bathe, and S. Maitra. Differential fault attack on kreyvium & flip. *IEEE Transactions on Computers*, 70(12):2161–2167, 2021. doi:10.1109/tc.2020.3038236.

[Rot76] O. S. Rothaus. On bent functions. *Journal of Combinatorial Theory, Series A*, 20:300–305, 1976. doi:10.1016/0097-3165(76)90024-8.

[SBM15] S. Sarkar, S. Banik, and S. Maitra. Differential fault attack against grain family with very few faults and minimal assumptions. *IEEE Transactions on Computers*, 64(6):1647–1657, 2015. doi:10.1109/tc.2014.2339854.

[SM01] P. Sarkar and S. Maitra. Efficient implementation of "large" stream cipher systems. In *CHES*, volume 2162 of *LNCS*, pages 319–332, 2001. doi:10.1007/3-540-44709-1_27.

[SM03] P. Sarkar and S. Maitra. Efficient implementation of cryptographically useful 'large' boolean functions. *IEEE Transactions on Computers*, 52(4):410–417, 2003. doi:10.1109/TC.2003.1190582.

[TKMM19] D. Tang, S. Kavut, B. Mandal, and S. Maitra. Modifying maiorana–mcfarland type bent functions for good cryptographic properties and efficient implementation. *SIAM Journal on Discrete Mathematics*, 33(1):238–256, 2019. doi:10.1137/18m1202864.

[TM18] D. Tang and S. Maitra. Constructions of n-variable ($n \equiv 2 \bmod 4$) balanced boolean functions with maximum absolute value in autocorrelation spectra $< 2^{n/2}$. *IEEE Transactions on Information Theory*, 64(1):393–402, 2018. doi:10.1109/tit.2017.2769092.

[TMM21] D. Tang, B. Mandal, and S. Maitra. Construction of balanced vectorial boolean functions with almost optimal nonlinearity and very low differential-linear uniformity. *Finite Fields and Their Applications*, 76:101903, 2021. doi:10.1016/j.ffa.2021.101903.